Ho Chi Minh city - University of Science
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Final Report

ADVANCED TEXT MINING

LECTURER

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Ho Chi Minh city, 2026

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1. Abstract

The rapid proliferation of digital information poses significant challenges in distinguishing between accurate and false information, with fake news potentially disrupting social order and threatening national security in Vietnam. While fact-checking is a rigorous process essential for making well-founded judgements, resources for low-resource languages like Vietnamese remain limited. In this study, we present VietFactCheck, an end-to-end system designed to support the verification of Vietnamese news by integrating Natural Language Processing (NLP) and machine learning techniques. Our system is primarily built upon the ViFactCheck dataset, the first multi-domain Vietnamese benchmark containing 7,232 human-annotated claim-evidence pairs across 12 diverse topics. To specifically bolster the Claim Detection stage, we introduced a novel dataset named ViNewsCheck, derived from ViFactCheck by grouping claims sharing the same context and utilizing the Gemini model to synthesize realistic news segments containing those claims for more robust experimentation. Unlike previous research that primarily focuses on the verification stage, we implement and evaluate a complete four-stage pipeline: Claim Detection, Document Retrieval, Evidence Selection, and Claim Verification. For the initial detection phase, we utilize Precision@K and chrF-based sentence alignment to ensure precise extraction of verifiable statements from unstructured text. For the subsequent retrieval and selection stages, we propose a hybrid mechanism combining SBERT embeddings, BM25, and TF-IDF. Our experimental results highlight the critical role of a Cross-Encoder Reranker in significantly enhancing performance during Evidence Selection compared to standard baseline methods. For Claim Verification, we prioritize practical deployment by fine-tuning Pre-trained Language Models (PLMs), such as PhoBERT and mBERT, ensuring high accuracy while remaining suitable for personal computing environments. The entire pipeline is rigorously evaluated using Accuracy, Macro F1-Score, and Macro F2-Score, demonstrating robust results across all components. To encourage further development and transparency in Vietnamese fact-checking, we have made our cleaned dataset, source code, and web application publicly available[1], with fine-tuned model weights hosted on the Hugging Face platform[2].

Keywords: Fact-checking, Vietnamese News, End-to-end Pipeline, Document Retrieval, Cross-Encoder Reranker, Pre-trained Language Models, Low-resource Language.

2. Introduction

Fact-checking has emerged as a multifaceted task at the intersection of information retrieval and natural language processing. It requires models to not only comprehend the semantics of a claim but also to reason over external evidence to verify its accuracy. This task has garnered significant global attention due to the rapid proliferation of digital information, where the spread of disinformation, rumors, and "fake news" poses severe threats to public discourse, social order, and national security. As noted by Lazer et al. (2018) [1], the extensive spread of false information can have profound negative impacts on individuals and society at large.

Over the years, numerous benchmark datasets have been developed to advance research in this domain. Notable among these is FEVER (2018) [2], which provided a massive set of claims sourced from Wikipedia to challenge models' understanding of textual verification. Subsequent efforts like FEVEROUS (2021) [3] expanded this to structured data , while MultiFC (2019) [4] and LIAR (2017) [5] introduced multi-domain claims and fine-grained labels from real-world fact-checking websites.

[1] *https://github.com/Namronaldo08102004/Vietfactcheck*
[2] *https://huggingface.co/collections/Namronaldo2004/vifactcheck-plm*

However, the vast majority of these resources are centered on the English language, leaving low-resource languages like Vietnamese underrepresented in the global landscape.

Recognizing this gap, research in Vietnam has begun to flourish. Initial efforts focused on Wikipedia-based textual knowledge, such as ViWikiFC (2024) [6], or utilizing Knowledge Graphs for verification as explored by Duong et al. (2023) [7]. Recently, the introduction of ViFactCheck [8] provided the first human-curated, multi-domain benchmark tailored specifically to Vietnamese online news, covering 12 diverse topics with over 7,200 claims. While ViFactCheck established a robust foundation for claim verification, existing evaluations often emphasize the final classification stage using "Gold Evidence," potentially overlooking the complexities of a real-world, end-to-end pipeline where evidence must first be discovered and filtered from massive, unstructured data sources.

In this study, we introduce VietFactCheck, an integrated fact-checking system designed to bridge the gap between static benchmarks and practical applications. Our motivation stems from the observation that a functional fact-checking tool requires a seamless transition through four critical stages: Claim Detection, Document Retrieval, Evidence Selection, and Claim Verification. Unlike previous works that focus on verification, our research dives deeper into the retrieval components. We implement a hybrid retrieval mechanism combining SBERT embeddings [9], BM25 [10], and TF-IDF to capture both semantic and lexical relevance. Furthermore, we introduce a Cross-Encoder Reranker to refine the retrieval process, demonstrating through rigorous experimentation that this extra step significantly improves the quality of evidence provided to the final classifier. To ensure our system is accessible for real-world use, our Claim Verification module utilizes fine-tuned Pre-trained Language Models (PLMs) - such as PhoBERT [11] - which are optimized for performance on personal computing environments without requiring massive hardware resources. We evaluate our pipeline using Accuracy and Macro F_1 Score, while introducing Macro F_2 Score for the Evidence Selection stage to prioritize the recall of relevant facts. For the Claim Detection module, we evaluate through Precision@K and chrF-based sentence alignment to ensure precise extraction of verifiable statements from unstructured text. Our complete system is deployed as a web application, allowing users to input news snippets and receive verified results with clear source references.

The remainder of this paper is structured to navigate the full development lifecycle of the VietFactCheck system. Following a review of related literature in Section 3, we describe our primary data foundations: the benchmark ViFactCheck dataset (Section 4) and our newly proposed ViNewsCheck dataset (Section 5), specifically designed for claim detection. Section 6 details the core architectural design of our four-stage pipeline. Comprehensive experimental results, performance analysis, and detailed discussions are presented in Section 7 . Finally, we demonstrate our practical web implementation in Section 8 before concluding with future research directions in Section 9.

3. Related Works

In this section, we provide an overview of existing datasets and methods related to automated fact-checking. We first examine representative datasets for fact-checking, including those in English and multilingual contexts, highlighting their key features, sizes, and limitations, particularly in terms of real-world applicability, evidence annotation, and reasoning complexity. We then discuss non-English datasets, with a focus on Vietnamese resources, to underscore the gaps in low-resource languages that the ViFactCheck dataset [8] addresses. Shifting to methods, we review prominent approaches aligned with our system's pipeline, covering the foundational steps of claim detection (or extraction), document retrieval, evidence selection, and claim verification. Emphasis is placed on theoretical foundations from reputable papers, including those in English and other languages, while incorporating Vietnamese-specific models where relevant to our implementation.

3.1. Fact-Checking Datasets

3.1.1. English and Multilingual Fact-Checking Datasets

The benchmark for automated fact-checking has evolved from simple classification to complex, multi-hop reasoning over heterogeneous data. The FEVER dataset [2] established the foundational pipeline by providing 185,445 claims mutated from Wikipedia. However, its primary limitation lies in the "artificiality" of claims, which often lack the linguistic ambiguity found in real-world misinformation. To address this, FEVEROUS [3] introduced a dataset of 87,026 claims requiring reasoning across both unstructured text and structured tables, forcing models to utilize cell-level retrieval and table-to-text alignment.

Moving toward real-world news, MultiFC [4] and LIAR [5] provide claims from fact-checkers like PolitiFact and Snopes. While LIAR focuses on veracity labels based on metadata, MultiFC offers 40 different label granularities and rich contextual information. Recent advancements have pushed toward specialized domains; for instance, SciFact [12] requires verifying scientific claims against citation graphs, and VitaminC [13] utilizes contrastive evidence (slight revisions in Wikipedia) to train models to be sensitive to factual updates. In the global context, X-Fact [14] spans 25 languages, revealing that multilingual models like mBERT and XLM-R still struggle with low-resource languages due to the lack of localized evidence. Most recently, FACTors [15] and Fact-Check Insights [16] have aggregated hundreds of thousands of fact-checks from IFCN-certified organizations, providing metadata-rich environments to study the temporal and political dynamics of misinformation.

3.1.2. Non-English and Vietnamese Fact-Checking Datasets

In non-English contexts, datasets like CHEF [17] (Chinese) and DANFEVER [18] (Danish) attempt to replicate the FEVER paradigm while incorporating cultural specificities. However, evidence scarcity remains a persistent challenge. For the Vietnamese language, a low-resource but linguistically complex language, the development of datasets is in a critical transition. ViWikiFC [6] initially provided a Wikipedia-based benchmark with over 20,000 claims, focusing on single-hop verification. However, Wikipedia's formal tone does not reflect the "noisy" nature of Vietnamese news or social media. ViNumFCR [19] recently introduced a dataset of 10,000+ social media claims focusing on numerical reasoning, revealing that standard encoders often fail at arithmetic inference (e.g., comparing percentages or dates). Specially, the dataset ViFactCheck [8] fills these gaps by providing 7,232 human-annotated claim-evidence pairs across 12 domains (health, politics, finance, etc.). Unlike ViWikiFC, ViFactCheck draws evidence from diverse Vietnamese news portals, emphasizing real-world applicability. It mandates multi-step reasoning - where a claim can only be verified by synthesizing information from multiple snippets - addressing the data sparsity and artificiality limitations of prior resources.

3.2. Fact-Checking Methods

Modern automated fact-checking systems have transitioned from monolithic classification models to modular, multi-stage pipelines. This architecture is designed to mimic the cognitive process of human fact-checkers: identifying a verifiable claim, gathering supporting or refuting evidence from trusted sources, and synthesizing a final verdict based on logical inference. The following sections provide a comprehensive technical review of the four primary components: claim detection, document retrieval, evidence selection, and claim verification, with a specific focus on the challenges and adaptations for the Vietnamese linguistic context.

3.2.1. Claim Detection and Check-worthiness

Claim detection is the foundational task of isolating "check-worthy" factual statements from a stream of unstructured text, such as political speeches, news articles, or social media feeds. This process involves a binary or multi-class classification problem where the system must distinguish between verifiable factual claims and non-claims (opinions, questions, or greetings).

Early research, such as ClaimBuster [20], focused on supervised learning with Support Vector Machines (SVMs) and Logistic Regression, relying heavily on hand-crafted features like TF-IDF, Part-of-Speech (POS) tags, and sentiment scores to rank sentences by their check-worthiness. However, these models struggled with the contextual ambiguity of natural language. The paradigm shifted with the introduction of BERT-based architectures [21], which utilize self-attention mechanisms to capture bidirectional context, making them the standard for binary claim classification. In multilingual and cross-lingual settings, XLM-R [22] and mBERT have been employed to transfer knowledge from high-resource languages (like English) to lower-resource ones.

For the Vietnamese language, additional complexities arise from its monosyllabic nature and the absence of clear word boundaries. PhoBERT [11], pre-trained on a 20GB corpus of Vietnamese text using a syllable-level approach, has shown a significant performance gain over multilingual counterparts. Recent advancements by Tran et al. (2023) [23] introduced a hybrid framework that augments PhoBERT with Knowledge Graphs (KGs). By linking extracted entities to a structured KG, the model becomes "entity-aware," allowing it to prioritize claims that involve critical legal entities or public figures, which is vital for the Vietnamese legal and news landscape.

3.2.2. Document Retrieval

Once a claim is identified, the system must fetch candidate evidence from a large-scale corpus (e.g., Wikipedia, official government portals, or news archives). This stage is often the bottleneck for recall in the entire pipeline. Traditional lexical methods like BM25 [10] utilize an improved version of TF-IDF to match keywords.5 While efficient, BM25 suffers from the vocabulary mismatch problem, where a claim and its supporting evidence use different words to describe the same concept (e.g., "văn bản pháp luật" vs. "thông tư/nghị định").

To overcome this, Dense Retrieval via Sentence-BERT (SBERT) [9] encodes both claims and documents into a shared d - dimensional latent space, where relevance is measured by cosine similarity:

$$\text{Score}(c, d) = \frac{v_c \cdot v_d}{\|v_c\| \|v_d\|} \tag{1}$$

BGE-M3 [24] has recently set a new benchmark in this domain by supporting multi-granularity (from short snippets to long documents) and multi-functionality (combining dense, sparse, and multi-vector retrieval).7 In the Vietnamese context, the Vietnamese_Embedding model [25], fine-tuned on specialized news triplets (Claim, Positive Evidence, Negative Evidence), provides high recall by understanding the formal syntax of Vietnamese journalism. State-of-the-art systems now increasingly adopt a Hybrid Search approach, which combines the exact-match capabilities of BM25 with the semantic depth of dense embeddings through techniques like Reciprocal Rank Fusion (RRF) [26]. RRF calculates a combined score based on the rank of a document in multiple retrieval lists (e.g., BM25 and BGE-M3) using the formula:

$$\text{Score}(d \in D) = \sum_{r \in R} \frac{1}{k + r(d)} \tag{2}$$

where $r(d)$ is the rank of document d in the r-th retriever and k is a constant (typically 60). This method is particularly effective for Vietnamese fact-checking because it does not require score normalization across different model scales, ensuring that high-precision lexical matches and high-recall semantic matches are both prioritized.

3.2.3. Evidence Selection

Retrieval typically returns a large set of candidate documents (e.g., Top-100), many of which are irrelevant "noise." Evidence Selection (or Reranking) refines this set to identify the specific sentences required for verification. This is often framed as a Cross-Encoder task. Unlike Bi-Encoders (used in retrieval), Cross-Encoders (e.g., Vietnamese_Reranker [27]) process the claim and evidence sentence jointly through the Transformer layers:

$$\text{Score} = \text{MLP}(\text{BERT}([\text{CLS}] + \text{Claim} + [\text{SEP}] + \text{Sentence})) \tag{3}$$

This allows for full self-attention interaction between every word in the claim and every word in the evidence, providing significantly higher precision at the cost of increased computation. For claims requiring multi-step reasoning, simple sentence-level filtering is insufficient. Graph-based architectures like GEAR [28] model evidence pieces as nodes in a graph, using Graph Attention Networks (GATs) to aggregate information across disparate documents. Similarly, the HOVER framework [29] emphasizes "many-hop" retrieval, where the model must identify "bridge" entities to find the final piece of evidence. Furthermore, the ERASER benchmark [30] has highlighted the necessity of extracting "rationales"—the minimum sufficient set of sentences—to ensure that the model's reasoning is interpretable and verifiable by human experts.

3.2.4. Claim Verification and Explainability

The final stage is the classification of the claim's veracity as SUPPORTS, REFUTES, or NOT ENOUGH INFO. This is technically modeled as a Natural Language Inference (NLI) or Recognizing Textual Entailment (RTE) task. Modern encoders like RoBERTa [31], when fine-tuned on large fact-checking datasets like FEVER, have reached human-level accuracy on synthetic data. However, Nie et al. (2019) [32] uncovered that these models often exploit "label leakage" (statistical artifacts in the dataset) rather than performing true logical reasoning.

The emergence of Large Language Models (LLMs) such as Gemma [33] and Llama 3 [34] has introduced a shift toward generative verification. By using Chain-of-Thought (CoT) prompting, these models can generate a natural language explanation that decomposes the reasoning process step-by-step before arriving at a verdict. Despite their power, LLMs are prone to "hallucinations"—generating factual-sounding but false information—particularly in low-resource languages like Vietnamese where pre-training data is less abundant.

To mitigate this, current state-of-the-art systems (including the one proposed in this work) implement a Retrieval-Augmented Generation (RAG) framework. In this setup, the LLM is strictly constrained to the evidence snippets provided by the reranker, effectively acting as a "reasoner" over external data rather than a "knowledge base." This grounding, combined with the survey insights from Kotonya and Toni (2020) [35], ensures that the system provides both a precise veracity label and a transparent, evidence-backed justification for its decision, which is critical for building user trust in sensitive domains like law and public health.

4. ViFactCheck Dataset

The ViFactCheck benchmark was developed to address the scarcity of guidance resources for analyzing the complex structure and semantics of the Vietnamese language in the context of misinformation. It comprises 7,232 human-annotated claim-evidence pairs sourced from reputable online news outlets. The claims and contexts are meticulously sourced from nine reputable, government-licensed Vietnamese online news outlets, ensuring high-quality linguistic content, contemporary relevance, and reliability. All content within the benchmark is entirely in Vietnamese, which allows for the formulation of claims and the extraction of evidence grounded deeply in the specific syntax and semantics of the native language.

By anchoring the dataset in multi-domain news - covering twelve diverse topics such as economics, education, and national security - it aims to bridge the gap between simple verification tasks and complex, real-world information scenarios in the context of Vietnam. The human-centered nature of the dataset construction introduces an additional layer of complexity; annotators were tasked with crafting intricate claims that often amalgamate multiple pieces of evidence and require multi-step reasoning. This design ensures high data quality and reliability, as evidenced by a Fleiss Kappa inter-annotator agreement score of 0.83, making ViFactCheck a valuable and robust resource for studying advanced inference in both pre-trained and large language models.

In this section, we begin by describing the data creation process of ViFactCheck in detail, which includes the three rigorous phases of data collection from official news sources, human-centered dataset annotation using specialized tools, and a multi-stage annotation validation process. Subsequently, we present an in-depth data analysis of the dataset, including basic statistical summaries of context and claim lengths, an exploration of the topic distribution, and a complexity analysis focused on reasoning steps and semantic similarity. Finally, we provide a comparative overview of ViFactCheck alongside other existing fact-checking benchmarks to highlight its unique contributions and standard-setting performance in the Vietnamese NLP landscape.

4.1. Data Creation

The construction of the ViFactCheck benchmark was executed through a systematic and rigorous three-phase methodology, designed to ensure both scientific depth and practical utility for Vietnamese natural language understanding. As illustrated in the comprehensive construction process pipeline shown in Figure 1, the workflow progressed through sequential stages of data collection, human-centered dataset annotation, and a multi-tiered annotation validation. Throughout each of these critical phases, the development was meticulously monitored and governed by experts to maintain the highest standards of dataset quality and linguistic integrity.

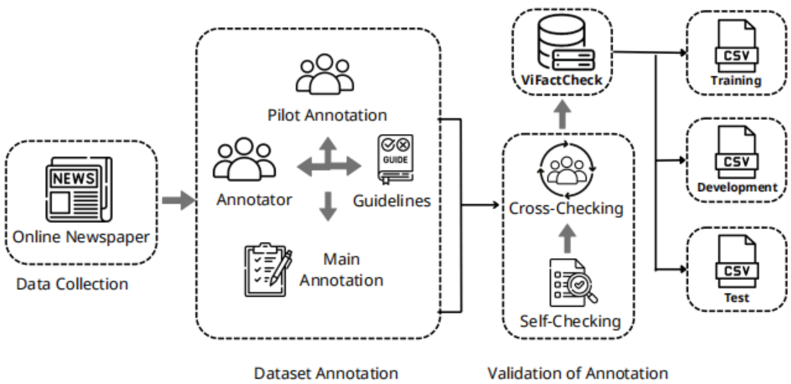


Figure 1: The ViFactCheck dataset contruction process.

4.1.1. Data Collection

The initial phase of the creation process involved the strategic selection and extraction of news content from the Vietnamese digital media landscape. To ensure the reliability and relevance of the information, the dataset was constructed using articles sourced from nine licensed and widely-read Vietnamese online newspapers. These specific sources - including Bao Chinh Phu, VnExpress, Dan Tri, Nguoi Lao Dong, Tuoi Tre, Tin Tuc, Phap Luat HCM, Thanh Nien, and Tien Phong - were chosen based on their government-licensed status, significant visitor counts, and their reputations for providing comprehensive and timely news coverage. Detailed information regarding the organizations and official URLs of these sources is presented in Table 1.

Website	Organization	URL
Bao Chinh Phu	Government of Vietnam	*https://baochinhphu.vn*
VnExpress	MOST Vietnam	*https://vnexpress.net*
Dan Tri	MOLISA Vietnam	*https://dantri.com.vn*
Nguoi Lao Dong	HCM City Committee	*https://nld.com.vn*
Tuoi Tre	HCM Communist Youth Union	*https://tuoitre.vn*
Tin Tuc	Vietnam News Agency	*https://baotintuc.vn*
Phap Luat HCM	HCM City People's Committee	*https://plo.vn*
Thanh Nien	Vietnam Youth Union	*https://thanhnien.vn*

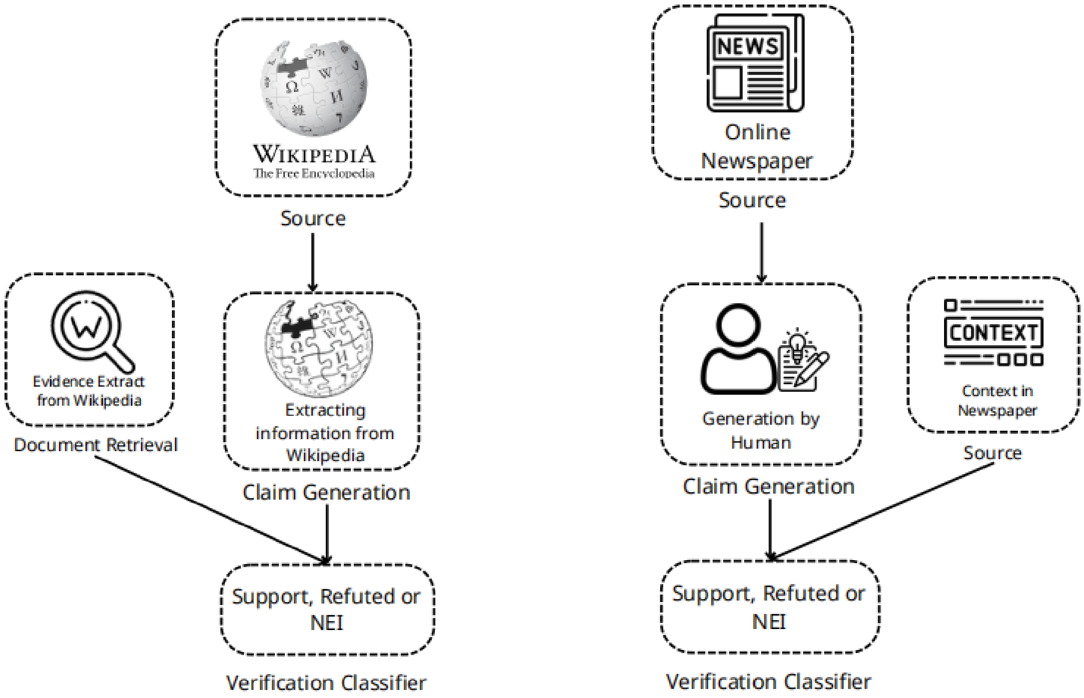
Table 1: Details of the sources and organizations of the online news sites in the ViFactCheck dataset.

Technically, the extraction process utilized two powerful Python libraries, BeautifulSoup and Selenium, which are well-regarded for their robust capabilities in scraping data from complex web structures. The collection effort specifically targeted articles published between February and March 2023. This specific timeframe was selected to capture the current dynamics of news reporting, thereby providing a contemporary snapshot of media trends and ensuring the dataset remains pertinent to modern misinformation patterns. From these sources, a wide array of metadata was extracted for each article, including the title, main content, topic categorization, lead descriptions, and source URLs.

The resulting initial corpus consisted of 1,000 articles spanning 12 diverse and popular topics frequently subject to public discourse in Vietnam. A critical methodological step during this phase was the creation of a "Full Context" field for each entry. This was achieved by merging the news leads - which typically summarize the most salient points - with their respective body contents. This enrichment provides a more comprehensive narrative view for each article, which is essential for facilitating the complex verification tasks and multi-step reasoning that the ViFactCheck dataset aims to evaluate.

4.1.2. Dataset Annotation

The annotation phase of the ViFactCheck dataset was built upon a human-centered methodology specifically designed to capture complex contextual nuances and factual details of online news. This approach intentionally deviates from conventional pipelines like FEVER, as contrasted in Figure 2. While traditional methods often rely on direct information extraction from sources like Wikipedia, the ViFactCheck process tasks human annotators with generating claims based on current news contexts to ensure the resulting data is representative of real-world information scenarios. By employing humans to interpret these nuances, the dataset is better equipped to support intricate inference tasks that require analysis across multiple pieces of evidence.



(a) Thorne et al. (2018) [2].

(b) Vifactcheck's process.

Figure 2: Comparison of the labeling pipelines in the FEVER and ViFactCheck datasets.

To facilitate this rigorous process, the developers recruited seven native Vietnamese-speaking university students, aged 20 to 22, representing diverse academic disciplines such as the social sciences, natural sciences, and Vietnamese studies. These annotators were selected based on their exceptional linguistic skills - evidenced by high scores in national Vietnamese literature exams - and their deep familiarity with various digital media platforms, ensuring the annotations reflect contemporary linguistic trends. Scientific rigor was further maintained through the continuous oversight of two native-speaking linguistic experts who established formal guidelines and monitored the entire annotation process. These experts provided a critical layer of depth to the methodology, leveraging their academic achievements to evaluate news content across various platforms.

The actual annotation work was conducted through a multi-stage process beginning with a pilot phase to ensure consistency. During this pilot, each annotator processed 120 claims derived from 20 random articles, receiving detailed feedback from experts to align their work with official requirements. Annotators were specifically instructed to proofread each claim carefully and adhere strictly to the guidelines. Following this familiarization, the main annotation phase commenced, where each participant was assigned a specific subset of data to ensure deep engagement and high-quality output.

For the technical implementation of the task, the team utilized Label Studio, an open-source platform providing an intuitive interface for complex labeling. The specific UI configured for this task, as illustrated in Figure 3, allowed annotators to manage titles, content, and claim generation fields effectively. Annotators were required to conduct a thorough review of each article before generating exactly six claim pairs per news item. This resulted in two pairs for each of the three defined labels: Support, Refute, and Not Enough Information (NEI). These labels were strictly defined to ensure grounding; Support and Refute claims were based on intrinsic information within the article, while NEI claims required the careful addition of external context that could not be definitively verified by the provided text.

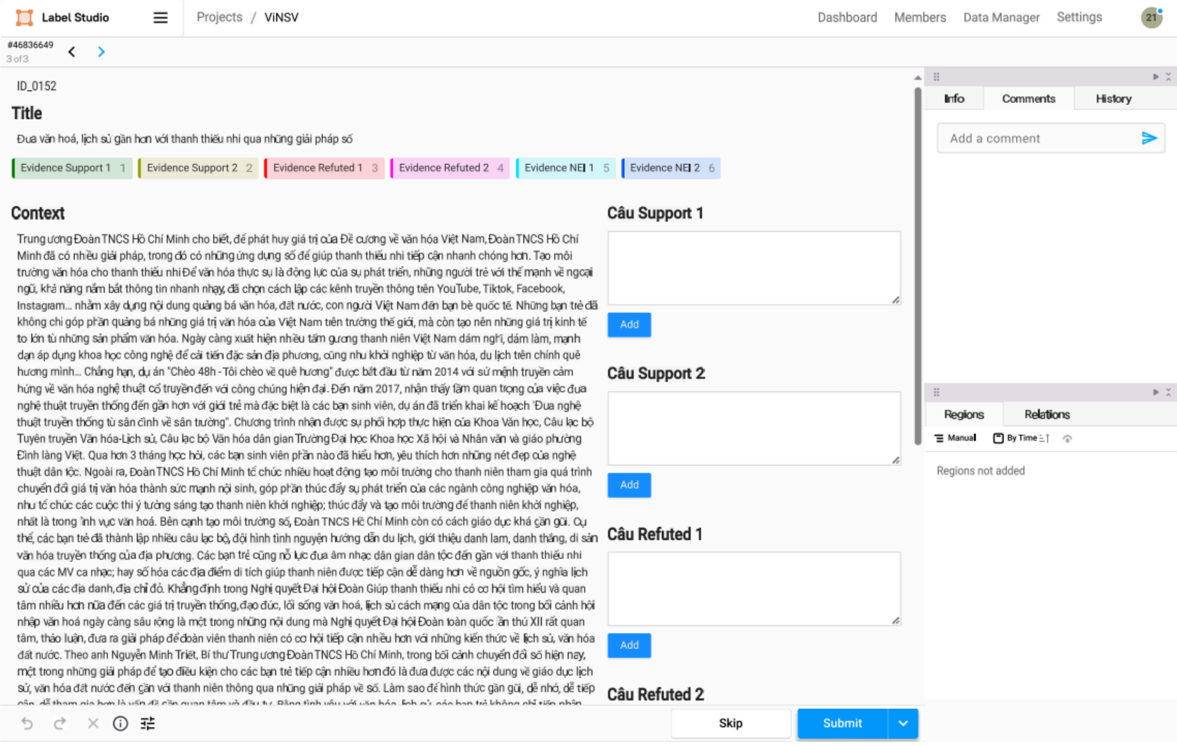


Figure 3: Label Studio UI for our annotation task.

The complexity of the dataset is further enriched by sophisticated human-generated rules summarized in Table 2. Annotators were encouraged to utilize their broad vocabulary and advanced sentence-writing techniques, moving beyond simple sentence identification to perform paraphrasing and semantic restructuring. Common strategies included "Restructuring the evidences" (used in 73.68% of support claims) and "Misjudging Event Dynamics" (used in 47.96% of refuted claims).

	Rules	Ratio (%)
Support	Restructuring the evidences	73.68
	Eliminating or adding words	44.21
	Substituting numbers, time, or mathematical inferences	7.34
	Altering the word order in a sentence	8.42
Refute	Employing Negation	8.16
	Replacing Words with Antonyms	17.35
	Misrepresenting quantity	22.45
	Misrepresenting Temporal Logic	16.37
	Misinterpreting Entity Relationships	5.11
	Misjudging Event Dynamics	47.96
NEI	Inferring sentences with unspecified information	90.20
	Utilizing external knowledge	10.78

Table 2: Approaches and rules for generating claims by humans in the ViFactCheck dataset. Note that a claim could involve multiple rules

While most claims relied on combining 1-2 of these rules, some annotators utilized four or more rules to create highly challenging data, as shown in the rule distribution in Figure 4. This diversity in claim generation forces models to perform genuine semantic inference rather than relying on simple keyword matching.

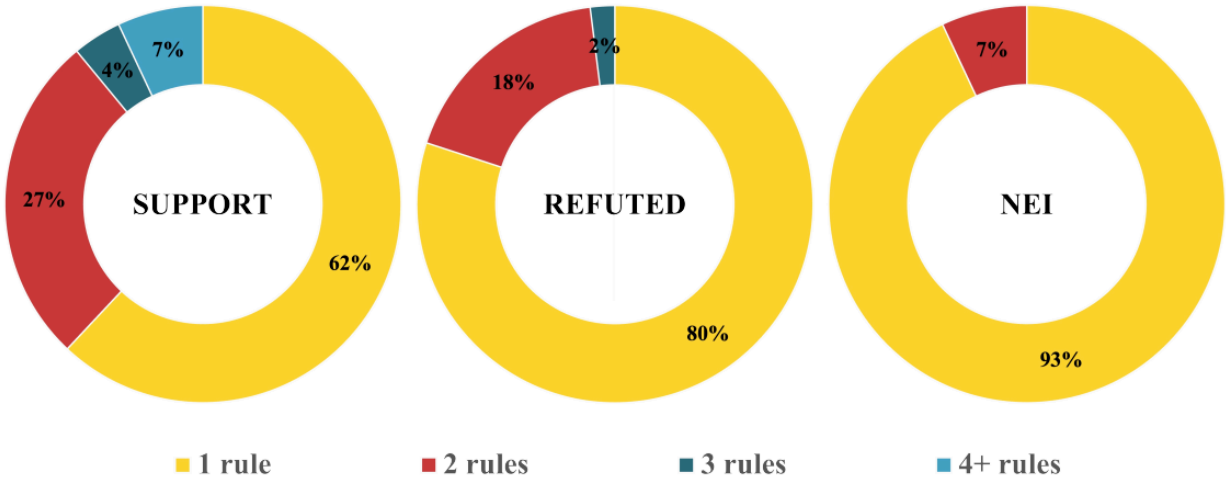


Figure 4: The ratio of combining different rules to create claims in ViFactCheck

As illustrated by the diverse samples presented in Table 3, the ViFactCheck dataset employs a sophisticated evidence-based framework for claim generation. Rather than relying on isolated statements, annotators were required to construct claims by synthesizing multiple informational cues, which are explicitly highlighted within the source text to provide a clear ground truth. This design choice deliberately elevates the reasoning difficulty, ensuring the benchmark mirrors the complexity of actual news verification while simultaneously bolstering its overall reliability. Ultimately, by anchoring the verification process in these verifiable evidence clusters, the dataset provides a rigorous training ground for language models to master the subtle semantic intricacies of the Vietnamese language.

Context	TPO - Tổng Công ty Cảng Hàng không Việt Nam (ACV) vừa chính thức gia hạn thời gian mời thầu thêm 1 tháng, kéo dài thời gian thực hiện gói thầu thi công nhà ga sân bay Long Thành từ 33 tháng lên 39 tháng. Như vậy, "siêu sân bay" Long Thành sẽ chỉ có thể đưa vào khai thác từ năm 2026 thay vì mục tiêu năm 2025 như trước đó. Tin từ ACV cho hay, đơn vị chính thức điều chỉnh kế hoạch và hồ sơ mời thầu gói thầu thi công xây dựng và lắp đặt thiết bị nhà ga hành khách sân bay Long Thành giai đoạn 1 (do ACV làm chủ đầu tư). Cụ thể, thời gian mời thầu được gia hạn thêm 1 tháng, kéo dài tới sáng ngày 28/4, thay vì tới ngày 28/3 như trước đó. ... Gói thầu thi công nhà ga hành khách sân bay Long Thành trị giá hơn 35 nghìn tỷ đồng do ACV làm chủ đầu tư. Đây là gói thầu lớn nhất dự án sân bay Long Thành... (**English**: TPO - Vietnam Airport Corporation (ACV) has officially extended the bidding period by an additional month, prolonging the implementation time for the construction contract of the Long Thanh Airport passenger terminal from 33 to 39 months. Consequently, the "mega airport" Long Thanh will only be operational by 2026 instead of the previous target of 2025. According to ACV, the organization has formally adjusted the plan and tender documents for the construction and installation of the passenger terminal at Long Thanh Airport Phase 1 (with ACV as the main investor). Specifically, the bidding period has been extended by one month, now ending on the morning of April 28, instead of the previous deadline of March 28. ... The construction contract for Long Thanh Airport's passenger terminal, valued at over 35 trillion VND is being managed by ACV. This is the largest contract within the Long Thanh Airport project...)
Support	Việc nhà thầu thi công xây dựng và lắp đặt thiết bị nhà ga hành khách sân bay Long Thành giai đoạn 1 bị điều chỉnh, thời gian bị kéo dài tới sáng ngày 28/4 thay vì tới ngày 28/3 như dự kiến. **English**: The construction and installation contract for the Long Thanh Airport Phase 1 passenger terminal has been adjusted, with the timeline extended to the morning of April 28 instead of the originally anticipated March 28.
Refute	Tổng Công ty Cảng Hàng không Việt Nam (ACV) vừa gia hạn thời gian mời thầu thêm thời gian 2 tháng, tức "siêu sân bay" Long Thành sẽ chỉ có thể đưa vào sử dụng từ năm 2026 thay vì năm 2025 như dự kiến ban đầu. **English**: Vietnam Airport Corporation (ACV) has recently extended the bidding period by an additional 2 months, meaning that the "mega airport" Long Thanh will only be operational by 2026 instead of the originally planned year 2025.
NEI	Gói thầu lớn nhất dự án sân bay Long Thành là gói thầu thi công nhà ga hành khách với trị giá hơn 35 nghìn tỷ đồng, được tài trợ bởi công ty Hàn Quốc. **English**: The largest contract within the Long Thanh Airport project is the construction of the passenger terminal, valued at over 35 trillion VND, and it is sponsored by a South Korean company.

Table 3: Typical samples from the **ViFactCheck** dataset with three labels **Support**, **Refute**, and **NEI**. The highlighted words is the evidence of the claim.

4.1.3. Validation of Annotation

Following the completion of the primary annotation phases, several systematic strategies were implemented to ensure the high quality, consistency, and reliability of the ViFactCheck dataset. The first stage of this verification process involved a self-checking phase, where annotators were required to perform a comprehensive review of their own generated claims and assigned labels. This internal audit focused primarily on identifying and correcting any potential grammatical errors or typographical mistakes that could impact the clarity of the data. Subsequently, a cross-checking mechanism was established to add an extra layer of objective scrutiny; in this stage, annotators verified the work of their peers, ensuring that the generated claim-evidence pairs adhered to the scientific standards defined in the guidelines. Any errors or inconsistencies identified during this peer review process were collaboratively discussed and corrected among the team members to reach a consensus.

To quantitatively evaluate the reliability of these human annotations, the development team utilized the Fleiss Kappa metric ([2], [36]), which serves as a widely recognized benchmark for measuring inter-annotator agreement (IAA) across various linguistic tasks. The evaluation was conducted by randomly selecting a significant sample of 10% of the total claims from the final labeled dataset, amounting to 726 entries. These samples were then assigned to a group of three annotators who were tasked with relabeling them independently. To prevent bias and ensure a rigorous assessment of the agreement, the existing annotations were hidden, requiring the evaluators to determine the labels without any knowledge of the original author's choices.

The analysis of this blind relabeling process yielded an inter-rater agreement score of 0.83. According to standard interpretations of the Fleiss Kappa coefficient, this result is indicative of a very high level of agreement among the annotators. Such a robust agreement level confirms that the guidelines were applied consistently across different individuals and highlights the overall high quality and reliability of the ViFactCheck benchmark.

4.2. Data Analysis

Following a rigorous construction process, the ViFactCheck dataset underwent a detailed Exploratory Data Analysis (EDA) to evaluate its linguistic characteristics, thematic scope, and the complexity of its reasoning tasks. These analyses aim to ensure that the dataset not only meets statistical standards but also accurately reflects the diversity and challenges inherent in the online news environment in Vietnam.

4.2.1. Basic Dataset Statistics

The ViFactCheck dataset consists of a total of 7,232 human-annotated claim-evidence pairs. The data is partitioned into training, development, and test sets with a split ratio of 7:1:2. Specifically, the training set includes 5,062 samples, the development set contains 723 samples, and the test set comprises 1,447 samples.

As detailed in the statistics in Table 4, the dataset demonstrates significant linguistic richness. The average length of a "Full Context" is approximately 700 words, with the longest context reaching 3,602 words. This contextual richness is particularly advantageous for large language models like Gemma, allowing them to extract maximum features from the data. Regarding the claims, each average claim contains about 36 words, with the most complex instances reaching up to 165 words. The total vocabulary size for the context portion of the training set is 25,382 words, providing a diverse linguistic framework for inference tasks.

		Training	Development	Test
Context	Total samples	1035	496	758
	Avg length	693.2	670.2	690.5
	Max length	3602	2534	3602
	Min length	71	71	71
	Total vocab size	25,382	16,522	21,263
Claim	Total samples	5062	723	1447
	Avg length	35.9	35.6	35.8
	Max length	165	145	135
	Min length	7	10	7
	Total vocab size	12,189	4,555	6,711

Table 4: Basic statistic of ViFactCheck dataset. The size and length of the vocab are computed at word level.

4.2.2. Topic and Domain Distribution

The ViFactCheck dataset covers 12 popular topics frequently found in Vietnamese online news, which are areas often prone to misinformation. The distribution of these topics, illustrated in Figure 5, shows a balanced approach across different social domains.

The "Headlines" topic, which includes updates on social issues and events, accounts for the largest proportion at 15.70%. Other significant domains contributing to the dataset include "Education" at 12.90%, "World" at 12.50%, and "Economics" at 10.90%. Notably, despite having a lower frequency of occurrence in the actual news cycle, the "National Security" topic was still included at a rate of 2.00%. The effort to collect news in this field stems from the urgent need for information accuracy regarding national security and public order.

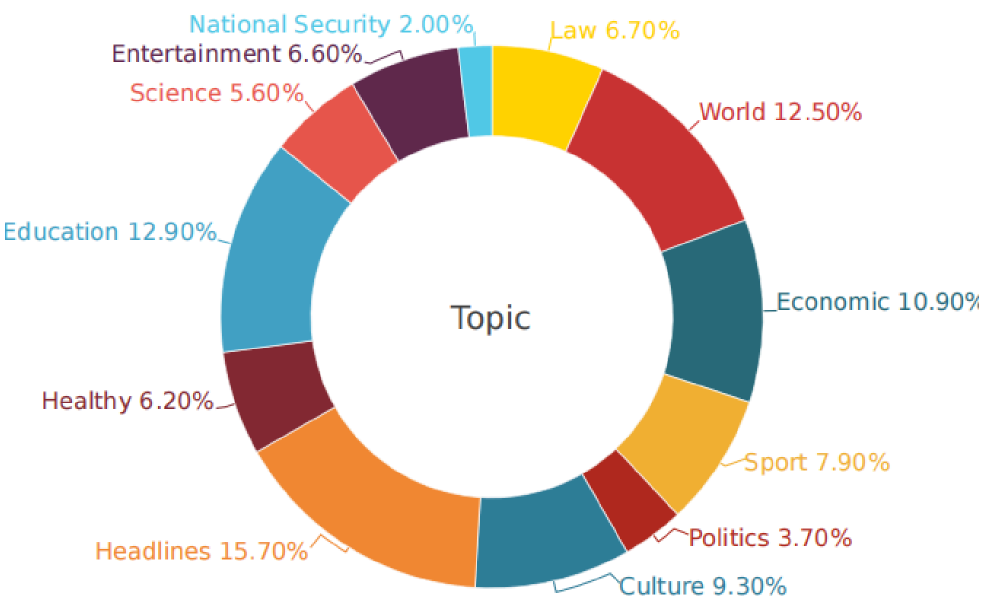


Figure 5: Topic distribution on ViFactCheck dataset.

4.2.3. Evidence Distribution and Reasoning Complexity

A hallmark of the ViFactCheck benchmark is its focus on multi-evidence reasoning, which goes beyond simple sentence-level verification tasks. According to the evidence distribution presented in Figure 6, while single-evidence claims predominate with 5,080 samples, the dataset still contains a large number of samples requiring the synthesis of information from multiple sources.

Specifically, there are 1,765 samples requiring two pieces of evidence and 293 samples requiring three pieces of evidence for verification. Furthermore, the dataset includes 130 samples where the claims require more than five separate pieces of evidence to reach a conclusion. This hierarchical structure requires models to possess advanced analytical skills to synthesize and validate information from long, unstructured texts. This strengthens the dataset's ability to train robust and multi-faceted fact-verification systems.

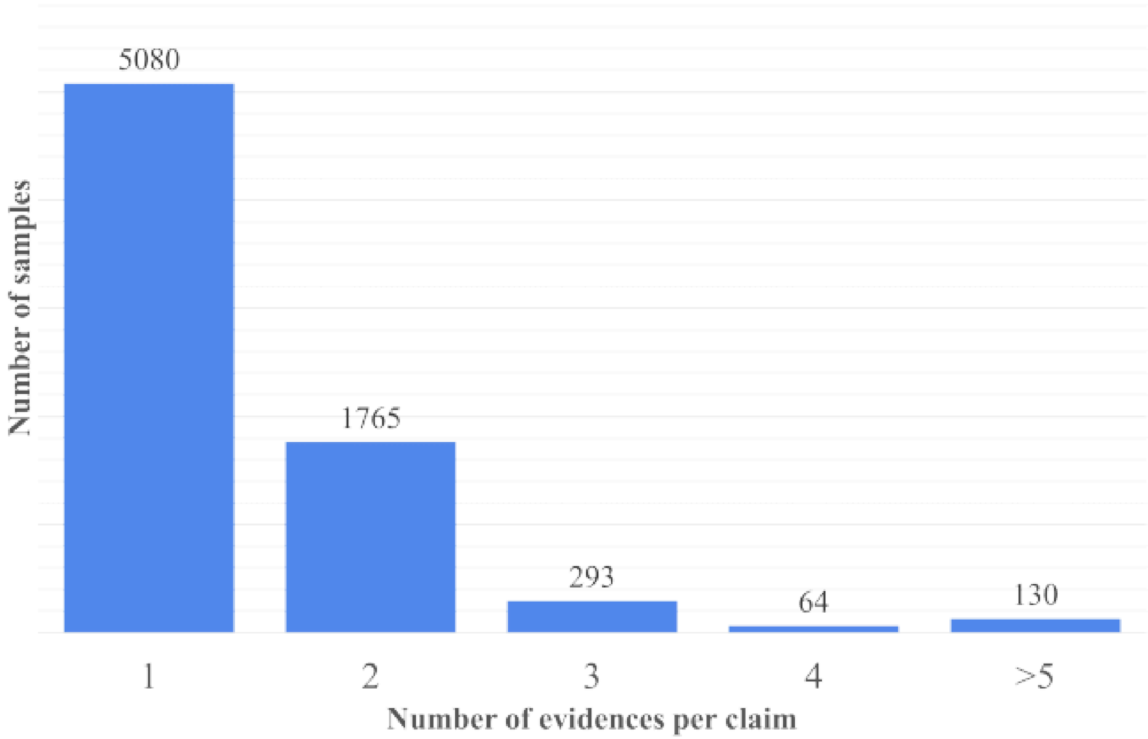


Figure 6: The distribution of single and multiple evidences samples in the ViFactCheck dataset.

4.2.4. Word Overlap and Semantic Similarity Analysis

To quantitatively assess the difficulty of inference tasks, the relationship between claims and their corresponding contexts was analyzed using word overlap and semantic similarity metrics. The results, summarized in Table 5, show that the dataset features claim-context pairs with minimal lexical overlap.

The metrics used include LCS (Longest Common Sequence), NWR (New Word Ratio), JS (Jaccard Similarity), and Lexical Overlap (LO). For example, "Support" labeled claims have a Jaccard Similarity (JS) of only 11.46% compared to the full context. This low overlap rate is a design intent to prevent models from relying on simple keyword matching or syntactic heuristics, forcing them to utilize advanced semantic reasoning capabilities. Conversely, there is a strong semantic correlation between claims and their "Gold Evidence," with "Related Words" (RW) scores reaching 86.89% for supported labels. This disparity emphasizes the importance of accurate evidence retrieval, as relevant information is often buried within a large context with low similarity.

		LSC	NWR	JS	LO	RW
Context	Support	20.60	6.54	11.46	20.13	36.24
	Refute	18.10	11.50	10.06	17.90	34.00
	NEI	19.89	11.81	10.96	18.50	32.85
Evidence	Support	17.70	17.13	63.52	73.87	86.89
	Refute	15.46	25.47	54.63	66.69	81.41
	NEI	16.71	26.84	57.56	64.39	81.13

Table 5: Relationship between claim-context and claimevidence in the ViFactCheck dataset.

4.2.5. Comparative Analysis of Fact-Checking Benchmarks

To provide a comprehensive view of the current state of automated fact-verification, it is essential to contextualize ViFactCheck within the broader global and local research landscape. Benchmark datasets serve as the vital foundation upon which verification algorithms are developed and fine-tuned, providing the diverse challenges necessary to advance the field. While English-language resources have traditionally dominated this domain, the shift toward multi-domain and multi-format datasets has highlighted the critical need for specialized benchmarks in low-resource languages. Table 6 provides a comparative overview of typical open-domain fact-checking datasets across various languages and sources.

	Dataset	Labels	Claims	Annotated Evidence	Language	Source	RS
English	FEVER (2018)	3	185,445	✓	English	Wikipedia	Multi
	FEVEROUS (2021)	3	87,026	✓	English	Wikipedia	Multi
	VitaminC (2021)	3	488,904	×	English	Wikipedia	Single
	MultiFC (2019)	2–40	36,534	✓	English	Fact-check	Multi
	LIAR (2017)	6	12,836	×	English	Fact-check	W/O
Non-English	CHEF (2022)	3	10,000	✓	Chinese	News/Fact-check	Multi
	DANFEVER (2021)	3	6,407	✓	Danish	Wikipedia	Multi
	ANT (2020)	2	4,547	×	Arabic	News	Multi
	ViWikiFC (2024)	3	20,976	×	Vietnamese	Wikipedia	Single
	ViFactCheck (2025)	3	7,232	✓	Vietnamese	News	Multi

Table 6: Comparative overview of typical open-domain fact-checking datasets. The type of Reasoning Steps (RS) column reflects the complexity involved in verifying the claims in each dataset.

As illustrated in Table 6, while prominent datasets like FEVER, LIAR, and the Vietnamese-specific ViWikiFC (2024) have advanced fact-checking research, they often remain limited by a reliance on Wikipedia, single-step reasoning, or a lack of annotated evidence. These constraints hinder their ability to capture the dynamic linguistic nuances of real-world digital media. ViFactCheck (2025) addresses these gaps by providing human-curated claims from nine licensed newspapers across 12 diverse topics. By incorporating multi-step reasoning and rigorously vetted human-annotated evidence, ViFactCheck offers a more robust and sophisticated foundation for evidence-based fact-checking in the Vietnamese context.

5. ViNewsCheck Dataset

The ViNewsCheck dataset is directly derived from the ViFactCheck benchmark, specifically designed to shift the focus from isolated claim-evidence pairs toward cohesive, multi-claim news narratives. The core of this derivation process involves aggregating individual human-annotated claims that share a common source article and utilizing the Gemini large language model to synthesize them into integrated news segments.

In this section, we provide a comprehensive overview of the development and structural characteristics of ViNewsCheck through two primary subsections. First, we detail the Data Creation process, which encompasses the systematic grouping of claims, the filtering criteria used to maintain dataset quality, and the specific prompt engineering strategies employed to generate the news segments. This is followed by an extensive Data Analysis that validates the robustness of the dataset. This analysis includes basic statistical summaries regarding the length and composition of the news segments, an examination of the diverse topic distribution, and a specialized study of the distribution of claims per news segment.

5.1. Data Creation

The development of the ViNewsCheck dataset begins with a systematic aggregation of claims derived from the ViFactCheck benchmark. We group individual claims based on their shared source URLs to ensure thematic consistency and maintain the original reporting context. To specifically target the challenges of multi-claim verification, a filtering threshold is applied: any article that yielded only a single claim in the original ViFactCheck benchmark is excluded from this version. This ensures that every entry in ViNewsCheck contains a minimum of two distinct verifiable statements, reflecting the complexity of real-world news. Following this filtering, we perform a normalization pass on the metadata to unify disparate topic labels into a standardized taxonomy, ensuring consistency across the twelve diverse domains.

Once the claims are grouped and pre-processed, we employ the Gemini large language model to synthesize these claims into a naturalistic news discourse. To achieve this, we utilize a specialized prompt that instructs the model to adopt the persona of an expert in fact-checking context creation. The synthesis process requires the model to generate a “fake context” that integrates the provided input statements verbatim while embedding them within logical transitions and a professional narrative structure. The specific prompt used for this generation, in both Vietnamese and its English equivalent, is as follows:

Vietnamese Prompt	**English Translation**
Bạn là chuyên gia tạo ngữ cảnh cho dữ liệu kiểm tra sự thật. Chủ đề: {topic} Cho các CÂU TUYÊN BỐ: {statements} Hãy tạo ra MỘT đoạn CONTEXT GIẢ sao cho: • Phù hợp với chủ đề • Logic, tự nhiên như bài báo • Sao chép nguyên văn statements nhưng bổ sung thêm các từ nối để câu văn trôi chảy, mạch lạc CHỈ TRẢ VỀ MỘT CHUỖI VĂN BẢN (string). KHÔNG JSON, KHÔNG markdown, KHÔNG giải thích.	You are an expert in creating context for fact-checking data. Topic: {topic}. For the STATEMENTS: {statements}. Please create ONE FAKE CONTEXT such that: • It fits the topic • It is logical and natural, like a news article • It copies the statements verbatim but adds transitions to make the text flow smoothly and coherently. ONLY RETURN ONE TEXT STRING. NO JSON, NO markdown, NO explanation.

This approach is designed to preserve the semantic integrity of the original human-annotated claims while providing a realistic, discursive environment for testing claim detection and verification models. By ensuring the claims are copied without modification into a fluid paragraph, we create a benchmark that tests a model's ability to distinguish fact-bearing sentences from surrounding transitionary text. A representative example of the resulting data can be seen in Table 7 below.

URL	*https://baotintuc.vn/the-gioi/trung-bay-ban-sao-bo-xuong-hoa-thach-cua-loai-khung-long-khong-lo-tai-london-20230330152500757.htm*
Topic	Thế giới (World)
Extracted Claims	**[Refute]**: Bộ xương bản sao được đưa đến Lodon bằng cách nghép thành con hoàn chỉnh rồi vận chuyển đến. (**English**: The replica skeleton was brought to London by being assembled into a complete creature and then transported.) **[Support]**: Bản sao bộ xương hóa thạch của một trong những loài khủng long lớn nhất trên Trái Đất lần đầu tiên trưng bày tại Bảo tàng Lịch sử tự nhiên ở thủ đô London. (**English**: The replica of the fossilized skeleton of one of the largest dinosaurs on Earth is being displayed for the first time at the Natural History Museum in London.) **[NEI]**: Bản sao của bộ xương hóa thách được tìm thấy sau khoảng 10 năm khai quật. (**English**: The replica of the fossilized skeleton was found after about 10 years of excavation.)
Synthesized News	Trong một sự kiện văn hóa thu hút sự chú ý của toàn thế giới, Bản sao bộ xương hóa thạch của một trong những loài khủng long lớn nhất trên Trái Đất lần đầu tiên trưng bày tại Bảo tàng Lịch sử tự nhiên ở thủ đô London. Theo thông tin từ ban tổ chức, đây là một thành tựu to lớn của giới khảo cổ học khi Bản sao của bộ xương hóa thách được tìm thấy sau khoảng 10 năm khai quật ròng rã tại Nam Mỹ. Để chuẩn bị cho buổi ra mắt hoành tráng này, quá trình hậu cần đã diễn ra vô cùng công phu, trong đó Bộ xương bản sao được đưa đến Lodon bằng cách nghép thành con hoàn chỉnh rồi vận chuyển đến sảnh chính của bảo tàng bằng chuyên cơ vận tải đặc biệt. (**English**: In a cultural event that has attracted global attention, the replica of the fossilized skeleton of one of the largest dinosaurs on Earth is being displayed for the first time at the Natural History Museum in London. According to information from the organizers, this is a major achievement for archaeology as the replica of the fossilized skeleton was found after approximately 10 years of excavation arduous excavation in South America. To prepare for this grand debut, the logistics process was extremely elaborate, in which the replica skeleton was brought to London by being assembled into a complete creature and then transported to the museum's main hall by a special transport aircraft.)

Table 7: A representative sample of the **ViNewsCheck** dataset illustrating the URL, thematic topic, original multi-label claims, and the integrated news context synthesized by Gemini.

5.2. Data Analysis

Upon completing the synthetic generation process, the ViNewsCheck dataset was subjected to a thorough Exploratory Data Analysis (EDA) to assess its structural properties, thematic coverage, and the distribution of claims within its synthesized narratives. In the following subsections, we delve into the statistical summaries, the claim distribution per news segment, and the topical diversity that define the dataset's unique characteristics as a benchmark for contemporary news verification.

5.2.1. Basis Dataset Statistics

The ViNewsCheck dataset comprises a total of 1,557 synthesized news segments, each containing a variable number of integrated, human-annotated claims. To facilitate effective model training and evaluation, the data is partitioned into training, development, and test sets. Specifically, the training set includes 1,010 samples, the development set contains 149 samples, and the test set comprises 398 samples.

As detailed in the statistics presented in Table 8, the dataset demonstrates substantial linguistic variety and narrative depth. In the training set, the average length of a synthesized news segment is approximately 224.09 words, with the most extensive context reaching 1,491 words. This structural density is particularly valuable for evaluating claim detection systems, as it requires models to identify verifiable statements within long-form, cohesive paragraphs rather than isolated sentences. Furthermore, the dataset features a significant vocabulary size, with 5,453 unique tokens in the training set and 3,191 in the test set, offering a rich linguistic framework for complex inference tasks across various domains.

	Training	**Development**	**Test**
Total samples	1010	149	398
Avg length	224.09	146.22	153.55
Max length	1491	427	574
Min length	55	58	79
Total vocab size	5,453	2,053	3,191

Table 8: Basic statistic of ViNewsCheck dataset. The size and length of the vocab are computed at word level.

5.2.2. Topic Distribution

To ensure that ViNewsCheck remains a robust and representative benchmark for real-world fact-checking, we analyzed the thematic scope of the dataset across twelve distinct domains. As illustrated in the topic distribution chart in Figure 7, the dataset demonstrates a wide-ranging coverage of contemporary issues in Vietnam and the world.

The most prominent theme is Culture & Society (Văn hóa - Xã hội), accounting for 14.84% of the total samples, followed closely by Education (Giáo dục) at 13.23% and World news (Thế giới) at 11.30%. Other significant categories include Economic (Kinh tế) at 9.12%, Sport (Thể thao) at 7.45%, and Law (Pháp luật) at 7.26%.

Less frequent but equally vital topics like Entertainment (Giải trí) at 4.82% and Politics (Chính trị) at 3.47% further enrich the dataset's diversity. This balanced topical distribution ensures that models evaluated on ViNewsCheck are tested against a variety of linguistic styles, terminologies, and subject-specific reasoning challenges.

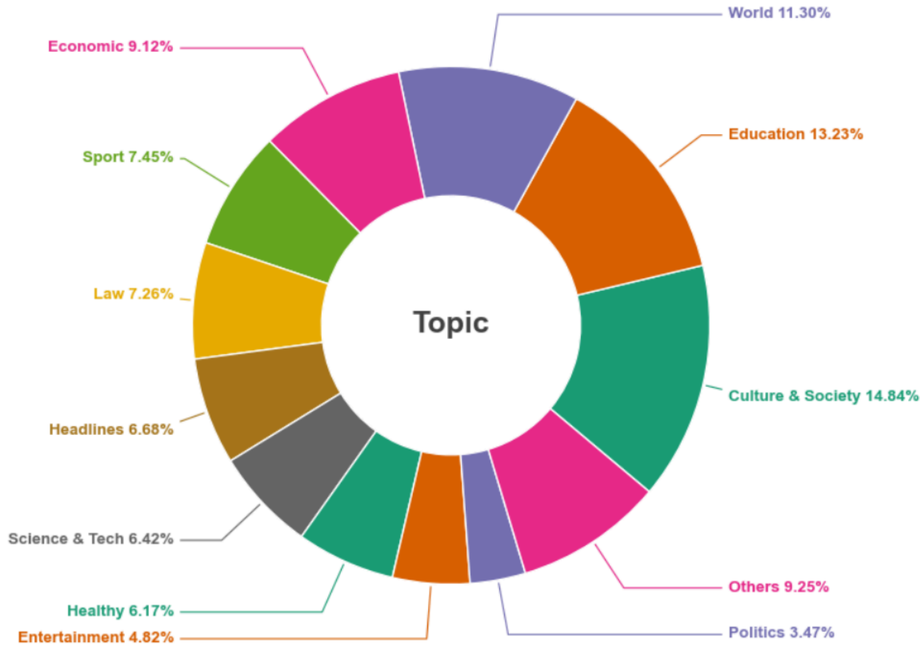


Figure 7: Topic distribution of ViNewsCheck dataset

5.2.3. Claim Quantity Distribution

A critical factor in assessing the complexity of the ViNewsCheck dataset is the density of verifiable claims embedded within each synthesized news narrative. As illustrated in the claim quantity distribution in Figure 8, the dataset encompasses a wide range of claim frequencies per news segment, directly aligning with our filtering requirement that each sample must contain at least two claims.

The distribution shows that the largest category consists of news segments with exactly two claims, totaling 438 samples. This is followed closely by segments containing three claims (331 samples) and four claims (325 samples). Even at higher levels of complexity, the dataset remains substantial, with 259 segments containing five claims and 204 segments featuring more than five claims integrated into a single cohesive discourse. This varied density ensures that ViNewsCheck serves as a robust benchmark for evaluating a model's ability to manage increasing levels of semantic density and perform sophisticated multi-claim verification within a natural news context.

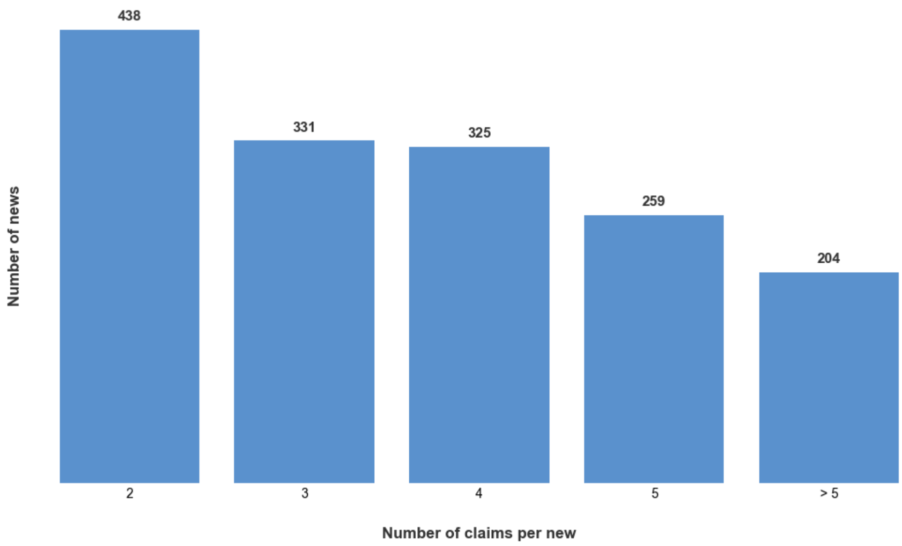


Figure 8: Frequency of news segments categorized by the number of embedded verifiable claims.

6. Our system design

This section details the architectural framework of VietFactCheck, an automated system designed to evaluate the authenticity of Vietnamese news through a multi-stage pipeline. Given the complexity of identifying misinformation, the system is engineered with a modular approach that integrates Natural Language Processing (NLP) and machine learning techniques to ensure both efficiency and accuracy. The architecture is designed to transform raw news inputs into verified claims supported by credible evidence, minimizing the spread of false information.

The proposed pipeline follows a sequential workflow consisting of four core components:

- **Claim Detection**: The initial stage responsible for isolating specific, verifiable statements from a larger body of text.
- **Document Retrieval**: A search mechanism that queries trusted databases to find relevant background information.
- **Evidence Selection**: A refining process that filters the retrieved documents to extract the most pertinent sentences or facts.
- **Claim Verification**: The final analytical module that classifies the claim as True, False, or Unknown based on the gathered evidence.

By organizing the system into these distinct modules, VietFactCheck provides a robust and scalable solution for real-time fact-checking. The following subsections provide an in-depth exploration of the methodology and technical implementation for each of these four stages.

6.1. Claim Detection

The Claim Detection module serves as the system's foundation, extracting verifiable factual statements to prevent error propagation in downstream tasks. To balance efficiency and complexity, we evaluate two complementary strategies: a Heuristic-based Extraction Module using Vietnamese syntactic rules and a Transformer-based Module utilizing BERTSum. This framework enables a direct comparison between rule-based interpretability and robust neural discourse modeling.

6.1.1. Heuristic based Claim Detection

To provide a lightweight and interpretable baseline for claim detection, we design a heuristic-based extraction module specifically tailored for the syntactic nuances of Vietnamese news articles. Unlike contemporary neural approaches that often function as "black boxes," this method formulates the claim detection task as a sentence-level classification problem grounded in linguistic rules derived from Vietnamese syntactic and semantic annotations. A significant advantage of this approach is that it does not rely on supervised training data, making it highly effective for scenarios where labeled datasets are scarce. Instead, the module leverages robust linguistic signals derived from part-of-speech (POS) tagging, named entity recognition (NER), and a set of handcrafted rules inspired by the common structural characteristics of factual statements in Vietnamese journalism.

6.1.1.1. Linguistic Analysis and Data Normalization

The primary objective of this module is to isolate declarative, fact-bearing sentences that are likely to represent verifiable claims. To facilitate this, the system utilizes the VnCoreNLP [37] framework to perform comprehensive Vietnamese-specific linguistic analysis. For every input document, the pipeline executes a multi-stage annotation process. This begins with word segmentation (wseg) to split the Vietnamese text into meaningful tokens - a crucial step given that Vietnamese is a monosyllabic language where spaces do not always denote word boundaries. This is followed by part-of-speech tagging (POS) to determine the grammatical roles of words and named entity recognition (NER) to identify critical information such as persons, organizations, and locations.

Because output formats can vary across different operating systems and library versions, we implemented a normalization procedure to unify various VnCoreNLP annotation structures into a consistent list-of-sentences representation. Within this framework, each sentence is stored as a sequence of token dictionaries containing word forms, POS tags, and NER labels. Furthermore, the extracted sentences undergo a refinement process to improve their readability as natural Vietnamese text. This involves replacing the underscores typically introduced by word segmentation (e.g., thành_phố becomes thành phố / “city”), correcting irregular spacing around punctuation marks, removing redundant whitespace, and standardizing dash formatting.

6.1.1.2. Heuristic Indicator Formulation

For a formal representation, we consider an input document $D = \{s_1, s_2, ..., s_N\}$, where each s_i represents a segmented sentence. Through the use of VnCoreNLP, we extract token-level representations $s_i = \{(w_{\mathrm{ij}}, p_{\mathrm{ij}}, n_{\mathrm{ij}})\}_{j=1}^{M_i}$, where w_{ij} is the word form, p_{ij} is the POS tag, and n_{ij} is the named entity label. To evaluate the “claim-likeness” of each sentence, we define three specific heuristic indicators. The first is the declarative indicator $h_1(s_i)$, which identifies whether a sentence follows the standard structure of a statement rather than a question or an exclamation. It is assigned a value of 1 if the sentence terminates with a period and contains no interrogative or exclamatory markers.

$$h_1(s_i) := \begin{cases} 1 & \text{if } s_i \text{ ends with "." and contains no "?" or "!"} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

The second heuristic is the factual verb indicator $h_2(s_i)$, which yields a value of 1 if the sentence contains at least one verb from a predefined factual verb set V_f. This set was constructed through a combination of linguistic prior knowledge and empirical inspection of Vietnamese news corpora. The set V_f includes essential verbs such as làm (do/make), theo (according to), ra (issue/produce), được (achieve/can), cần (need), có (have/possess), cho (give/for), phát triển (develop), lên (increase/up), tổ chức (organize), bị (passive marker/suffer), phải (must/have to), thực hiện (implement/carry out), and là (be/is/are). This indicator is formally defined as:

$$h_2(s_i) := \begin{cases} 1 & \text{if } \exists j : p_{\mathrm{ij}} \in V^* \wedge w_{\mathrm{ij}} \in V_f \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

In this equation, V^* represents the set of valid Vietnamese verb POS tags. Finally, we utilize an entity-or-number indicator $h_3(s_i)$, which assumes that a verifiable claim often contains specific subjects or quantitative data. This indicator is triggered if the sentence includes either a named entity or a numerical token, representing the presence of concrete evidence.

$$h_3(s_i) := \begin{cases} 1 & \text{if } \exists j : n_{\mathrm{ij}} \neq O \vee n_{\mathrm{ij}} = M \\ 0 & \text{otherwise} \end{cases} \tag{6}$$

Here, $n_{\mathrm{ij}} \neq O$ signifies the presence of a recognized named entity, while $n_{\mathrm{ij}} = M$ identifies tokens categorized as numbers or quantities.

6.1.1.3. Synthesis and Sentence Classification

To determine the final classification of a sentence, these three indicators are aggregated through an additive scoring function. This function produces a total score $S(s_i)$ within the range $[0, 3]$, reflecting the cumulative strength of the linguistic signals present in the sentence.

$$S(s_i) = h_1(s_i) + h_2(s_i) + h_3(s_i) \quad (7)$$

A sentence is formally classified as a claim only when it satisfies a majority of the heuristic criteria, specifically when $S(s_i) > 2$. This threshold ensures that the predicted claim set C consists of sentences that are simultaneously declarative, contain factual action verbs, and mention specific entities or figures.

$$C = \{s_i \in D \mid S(s_i) > 2\} \quad (8)$$

By executing this extraction process once per document, the module achieves the predicted claim set C_D with high computational efficiency, providing a transparent and reproducible foundation for further analysis.

6.1.2. Transformer based Claim Detection

To complement the interpretable heuristic approach, we implement a neural claim detection module based on the BERTSum [38] framework. This architecture treats claim detection as an extractive task, leveraging the power of pre-trained BERT encoders to identify claim-worthy sentences within Vietnamese news documents. For a given input document $D = \{s_1, s_2, ..., s_n\}$ consisting of n individual sentences, the system first transforms the raw text into a high-dimensional feature space. BERTSum utilizes a BERT-based contextual encoder to process each sentence, generating sentence-level representations $h_i \in \mathbb{R}^d$. These vectors encapsulate the semantic essence of each sentence within its local context.

Following the initial encoding, these representations are passed through an inter-sentence Transformer layer. This specific component is critical as it models global dependencies across the entire document, rather than treating sentences in isolation. By applying self-attention mechanisms at the discourse level, the system can capture the relative importance of each sentence and its contextual salience compared to the rest of the text. This allows the model to differentiate between background information and the core, verifiable claims that constitute the focus of the fact-checking pipeline. Once the global features are refined, a sigmoid-based classifier is employed to assign a probability-based extraction score p_i to each sentence.

$$p_i = \sigma(W h_i + b) \quad (9)$$

In this formulation, $\sigma(\cdot)$ denotes the sigmoid activation function, while W and b represent the learnable weight matrix and bias vector, respectively. Sentences that yield the highest scores are prioritized and selected as the predicted claims, ultimately forming the output set $C = \{c_1, c_2, ..., c_k\}$.

In our specific implementation, the BERTSum module is executed in batch mode to optimize computational performance and resource management. Rather than processing documents sequentially, the system aggregates all valid document contexts and passes them to the model simultaneously. This design choice ensures that the heavy transformer model is loaded into memory only once, with inference performed through a single forward pass over the entire dataset. By avoiding the overhead associated with redundant model initializations, this batch processing strategy substantially reduces the total runtime. Formally, for a collection of M documents $\{D^{(1)}, ..., D^{\{((M))\}}$, the system generates the corresponding claim sets $\left\{C^{(1)}, ..., C^{(M)}\right\}$ in a highly efficient, unified operation.

The strength of this Transformer-based approach lies in its ability to learn implicit syntactic and semantic patterns directly from the data. Unlike the heuristic-based method, which is restricted by explicit linguistic rules, BERTSum utilizes its self-attention layers to capture long-range dependencies and complex discourse structures. This data-driven nature makes the model significantly more robust to lexical variations and intricate sentence constructions commonly found in modern Vietnamese journalism. By integrating both the interpretable heuristic method and this strong neural baseline, the proposed system enables a rigorous comparative analysis between rule-based and deep learning approaches in the domain of Vietnamese claim detection.

6.2. Document Retrieval and Evidence Selection

This section provides a comprehensive description of the Document Retrieval and Evidence Selection modules. These stages form a hierarchical filtering pipeline designed to navigate the large search space of Vietnamese news and identify the precise sentences required for claim verification. Our approach combines traditional information retrieval (IR) methods with state-of-the-art deep learning reranking to balance computational efficiency with semantic precision.

6.2.1. Hybrid Retrieval Framework

As illustrated in the system architecture in Figure 9, our retrieval engine processes a claim and a full-context corpus through a multi-stage pipeline. The process begins with Document Indexing, where the corpus is segmented into individual sentences using the underthesea library for accurate Vietnamese tokenization. To handle the semantic and lexical nuances of the Vietnamese language, we implement a hybrid scoring mechanism that integrates three distinct retrieval models:

- **BM25L (Best Matching 25 - Length Normalized)**: An evolution of BM25 that better handles document length variations by calculating lexical relevance based on term frequency and inverse document frequency.
- **TF-IDF (Term Frequency-Inverse Document Frequency)**: A classic statistical measure used to evaluate the importance of a word within a document relative to the entire collection.
- **SBERT (Sentence-BERT)**: A semantic embedding model (utilizing the Vietnamese Reranker of AITeamVN for encoding) that captures the underlying meaning of sentences in a high-dimensional vector space.

For any given query (claim) q and a candidate sentence s, we calculate a hybrid score. Since the raw scores from these three models operate on different scales, we first apply Min-Max Normalization to map each score into the range $[0, 1]$:

$$S_{\text{norm}} = \frac{S - S_{\text{min}}}{S_{\text{max}} - S_{\text{min}}} \tag{10}$$

The final ranking score S_{hybrid} is then computed as a weighted linear combination:

$$S_{\text{hybrid}} = w_1 \cdot S_{\text{BM25L}} + w_2 \cdot S_{\text{TF-IDF}} + w_3 \cdot S_{\text{SBERT}} \tag{11}$$

Based on empirical tuning, our system defaults to weights of $(0.3, 0.3, 0.4)$ for broad retrieval and $(0.2, 0.2, 0.6)$ for fine-grained evidence selection, emphasizing semantic similarity in the latter stage.

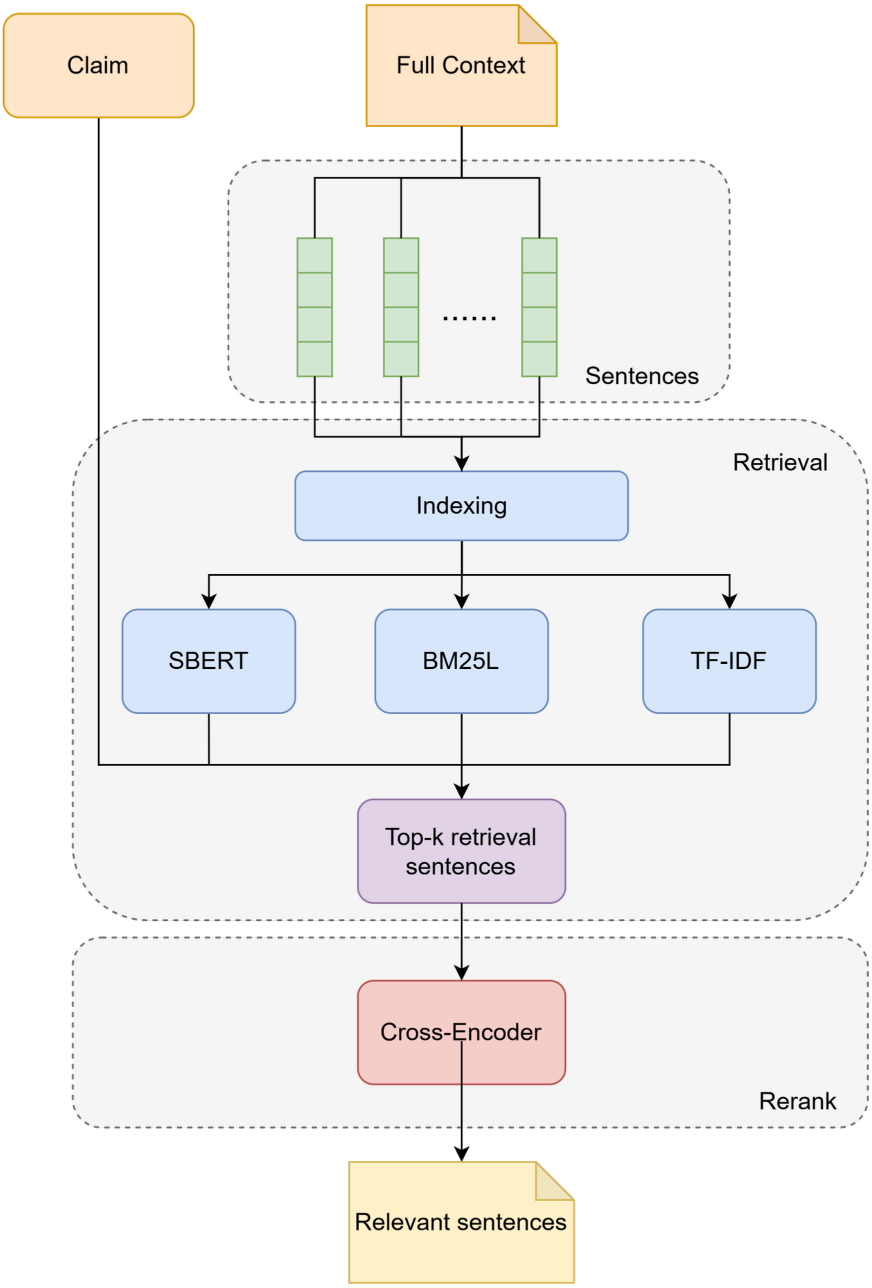


Figure 9: Proposed architecture of the hybrid Document Retrieval and Evidence Selection pipeline.

6.2.2. Document Retrieval (URL-level)

The first phase of the pipeline is Document Retrieval, which aims to identify the specific news article (source) that serves as the context for the claim. Given that our dataset focuses on claims derived from single articles, we utilize the source URL as a unique identifier for deduplication and indexing.

- **Candidate Generation**: The system computes the hybrid scores for all sentences in the indexed corpus.
- **Aggregation**: Scores are aggregated by their parent URL. For each URL, we retain the maximum sentence score as the representative relevance score for that article.
- **URL Selection**: The top k URLs (default $k = 3$) are selected. To ensure the highest accuracy, these candidates are passed to the Vietnamese Reranker (a Cross-Encoder model). The Cross-Encoder computes a deep interaction score by processing the claim and the full article context together, selecting the single "Best URL" as the final context source.

6.2.3. Evidence Selection (Sentence-level)

Once the relevant article is identified, the Evidence Selection module must extract the specific sentences that provide factual support or contradiction for the claim. This task is challenging because the number of necessary evidences varies significantly across different claims. As shown in the statistical distribution of the ViFactCheck dataset in Figure 6, most claims rely on a single evidence sentence (5,080 samples), while a significant portion requires two (1,765 samples), and a smaller fraction requires up to seven or more.

To handle this variability, we implement a **Dynamic Selection Algorithm** instead of using a fixed k. The selection logic operates in three steps:

- **Top-k Filtering**: Within the identified "Best URL", we retrieve a candidate set $\mathcal{S} = \{s_1, s_2, ..., s_k\}$ using the hybrid scoring method. In our system, we set $k = 10$ as the default value.
- **Cross-Encoder Reranking**: Candidates are reranked using the Vietnamese Reranker to obtain a precision score $R(s_i)$ for each sentence.
- **Hierarchical Selection Logic**: We apply a dual-threshold approach to define the final evidence set E:
  - **High Confidence Selection**: Any sentence with a rerank score exceeding a primary threshold T_1 (default 0.75) is automatically included in the evidence set:

$$E = \{s_i \in \mathcal{S} \mid R(s_i) \geq T_1\} \tag{12}$$

  - **Relative Gap Selection**: If $E = \emptyset$, the top-ranked sentence s_1 is selected as the anchor. Subsequent sentences are included based on the relative score degradation between consecutive candidates:

$$s_i \in E \Leftrightarrow R(s_{i-1}) - R(s_i) < T_2 \tag{13}$$

  where T_2 is the gap threshold (default 0.05).

This adaptive strategy allows the system to be flexible when multiple sentences contribute to a claim (capturing the varied evidence distribution seen in Figure 6) while remaining concise when a single sentence is sufficient, effectively filtering out noise that might confuse the final verification model.

6.3. Claim Verification

The final component of the VietFactCheck pipeline is the Claim Verification module. This stage is responsible for determining the veracity of a claim based on the information retrieved during the evidence selection phase. Formally, this is treated as a sequence classification task where the system assigns one of three logical labels to each claim: Supported (the claim is confirmed by the evidence), Refuted (the claim is contradicted by the evidence), or Not Enough Information (NEI) (the evidence is insufficient to verify or refute the claim).

6.3.1. Model Architectures and Baseline Selection

To achieve robust performance in the Vietnamese context while ensuring the system can operate efficiently on standard personal computers, we utilize various Pre-trained Language Models (PLMs) based on the Transformer architecture . These models leverage deep learning to process and analyze contextual patterns, making them highly effective for evidence-based verification. Following the benchmarks established in the ViFactCheck study, our module incorporates several state-of-the-art PLMs, categorized by their linguistic focus:

- **Monolingual Models**: These models are specifically tailored to handle the unique linguistic nuances of the Vietnamese language. PhoBERT leverages the RoBERTa architecture and has been shown to discern subtle language patterns that multilingual models might overlook. ViBERT, also based on the BERT architecture, is pre-trained on a substantial 10GB corpus of uncompressed Vietnamese text to achieve optimal monolingual performance.
- **Multilingual Models**: These models provide a broader cross-lingual context by being trained on vast datasets across dozens of languages. mBERT is trained on 104 languages, including Vietnamese, facilitating comprehensive linguistic analysis. XLM-R (base and large) expands this scope to 100 languages, allowing the system to potentially leverage cross-lingual information.

The technical specifications of these baseline models - including the number of hidden layers, attention heads, total parameters, and maximum sequence length - are summarized in Table 9 (derived from the specifications in the benchmark paper).

6.3.2. Input Processing and Context Strategies

The verification process relies on the interaction between the claim and the provided context. Our system is designed to be flexible, supporting two primary modes of context input: **Full-context** and **Evidence-based**.

- **Full-context Mode**: The claim is evaluated against the entire content of the news article. While this provides a comprehensive narrative view, it often introduces extraneous information that can act as noise during the classification process.
- **Evidence-based Mode**: The system focuses exclusively on the relevant sentences extracted by the retrieval module. As demonstrated in the ViFactCheck benchmark, using gold evidence typically leads to significantly higher accuracy scores. By isolating directly relevant snippets, the model can concentrate its computational resources on pertinent facts, thereby reducing noise and improving precision.

For input preparation, the claim and its associated context are concatenated into a single input pair using the AutoTokenizer. In our implementation, we employ the encode plus method with a fixed max length of 256, ensuring truncation and padding for consistent batch processing. For monolingual models like PhoBERT, we apply specialized word segmentation using VnCoreNLP to ensure the input tokens align with the linguistic requirements of the model.

Model	**#Layer**	**#Head**	**#Params**	**#Vocab**	**#MSL**	**Domain data**	**Language support**
PhoBERT$_{base}$	12	12	135M	64K	256	ViWiki + ViNews	Vietnamese
PhoBERT$_{large}$	24	16	370M	64K	256	ViWiki + ViNews	Vietnamese
ViBERT	12	12	-	30K	256	Vietnamese News	Vietnamese
mBERT	12	12	110M	30K	512	Wikipedia + BookCorpus	Multilingual
XLM-R$_{base}$	12	12	270M	250K	512	CommonCrawl	100+ languages
XLM-R$_{large}$	24	16	550M	250K	512	CommonCrawl	100+ languages

Table 9: Detailed specifications of our baseline models. Abbreviations used are: Layers (Number of Hidden Layers), Heads (Number of Attention Heads), Params (Total Number of Parameters), Vocab (Vocabulary Size), and MSL (Maximum Sequence Length).

6.3.3. Experimental Setup and Hyperparameters

To ensure the models are properly adapted to the fact-checking task and to maintain consistency with the results reported in the ViFactCheck dataset, we conduct supervised fine-tuning. This process involves adjusting the model weights to recognize the specific semantic relationships between Vietnamese claims and supporting evidence. The specific hyperparameter configurations used to fine-tune the PLM baselines are presented in Table 10.

Hyperparameters	**Value**
Epochs	10
Batch Size	16
Optimizer	AdamW
Learning Rate	5e-6
Dropout Rate	0.3

Table 10: Hyperparameter settings for fine-tuning pre-trained language models.

7. Experiments

This section presents the experimental evaluation of the four key stages in our proposed fact-checking pipeline: Claim Detection, Document Retrieval, Evidence Selection, and Claim Verification. To rigorously assess the performance of each component, we employ a diverse set of metrics tailored to their specific objectives. For the initial Claim Detection module, performance is measured using Precision@K in conjunction with chrF-based sentence alignment to account for linguistic and structural variations. The subsequent Document Retrieval, Evidence Selection, and Claim Verification stages are evaluated using standard classification metrics, specifically Accuracy and the F_β Score. This section concludes with a comprehensive Discussion focused on how external factors - including context length, evidence quantity, and the selection of top-k values - influence the overall efficacy and robustness of the fact-checking process.

7.1. Evaluation Metrics

This section defines the evaluation framework used to assess the performance of the VietFactCheck system across its four primary components: Claim Detection, Document Retrieval, Evidence Selection, and Claim Verification. To provide a task-appropriate assessment for these heterogeneous subtasks, we categorize our evaluation criteria into two distinct groups. The following subsections detail the specific metrics employed for evaluating the initial extraction phase and those applied to the subsequent classification-driven components.

7.1.1. Claim Detection Metrics

This section describes the evaluation methodology used to assess the effectiveness of the proposed claim detection module. Since claim detection is formulated as an extractive sentence selection problem, conventional classification metrics such as accuracy or F1-score are insufficient to capture partial matches and semantic overlap between predicted claims and reference statements. Instead, we adopt retrieval-oriented metrics combined with character-level similarity measures to provide a more flexible and fine-grained evaluation. Specifically, the system performance is measured using **Precision@K** together with **chrF**-based sentence alignment, allowing us to quantify both ranking quality and textual similarity between extracted claims and gold statements.

Let $D = \{s_1, s_2, ..., s_n\}$ denote a document consisting of n sentences, and let $C = \{c_1, c_2, ..., c_m\}$ represent the set of claims extracted by the model, ordered by their predicted importance. For each gold statement g, all sentences in D are ranked according to their character n-gram F-score (chrF) with respect to g as display in Equation 14, producing a pseudo-reference set G_K consisting of the top-K most similar sentences. This procedure approximates the absence of explicit sentence-level annotations by identifying the most relevant contextual sentences for each claim.

$$\text{chrF}\beta = (1+\beta^2)\frac{\text{chrP}\cdot\text{chrR}}{\beta^2\cdot\text{chrP}+\text{chrR}} \tag{14}$$

In this formulation, chrP and chrR represent the character n-gram precision and recall, respectively, both of which are arithmetically averaged over all considered n-gram lengths. Specifically, chrP denotes the percentage of n-grams in the hypothesis that have a counterpart in the reference, while chrR signifies the percentage of character n-grams in the reference also present in the hypothesis. For the evaluation within the VietFactCheck framework, we adopt standard values of 1 for β and 4 for the n-gram length.

Precision at K is then computed as:

$$\text{Precision@K} = \frac{1}{K}\sum_{\text{i=1}}^{K} I(c_i \in G_K), \tag{15}$$

where $I(c\,\cdot)$ is an indicator function equal to 1 if the predicted claim c_i belongs to the chrF-derived reference set G_K, and 0 otherwise. Precision@K is calculated for multiple values of K (namely $K = \{1, 3, 5\}$) to evaluate the quality of the top-ranked extracted claims.

To obtain dataset-level performance, Precision@K scores are averaged across all gold statements rather than documents, yielding a statement-centric evaluation that reflects retrieval effectiveness at fine granularity. In addition, the average number of extracted claims per document is reported to characterize model behavior and extraction density. This combined evaluation framework enables fair comparison between heuristic-based and Transformer-based approaches while accommodating paraphrasing and lexical variation commonly observed in Vietnamese news text.

7.1.2. Accuracy-based Metrics

This subsection describes the quantitative metrics used to evaluate the VietFactCheck pipeline across the Document Retrieval, Evidence Selection, and Claim Verification modules. These metrics are selected to reflect the system's overall accuracy and its ability to handle the complexities of Vietnamese news verification.

Accuracy is utilized as a fundamental metric to measure the overall proportion of correct predictions across the dataset. It is defined as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \tag{16}$$

To gain a more granular understanding of model performance, we utilize the F_β Score, which evaluates the trade-off between Precision (P) and Recall (R). Precision measures the accuracy of positive predictions, while Recall measures the ability to capture all relevant instances. These are defined as:

$$\text{P} = \frac{\text{TP}}{\text{TP}+\text{FP}}, \qquad \text{R} = \frac{\text{TP}}{\text{TP}+\text{FN}} \tag{17}$$

The general F_β Score is the weighted harmonic mean of precision and recall, formulated as:

$$F_\beta = (1+\beta^2) \cdot \frac{P \cdot R}{(\beta^2 \cdot P) + R} \tag{18}$$

In our experimental setup, we apply two specific configurations of this framework:

- F_1 Score ($\beta = 1$): This metric provides a balanced weight to both precision and recall, and is used for evaluating the **Document Retrieval** and **Claim Verification** modules.
- F_2 Score ($\beta = 2$): This metric weights recall more heavily than precision. We specifically apply the F_2 Score to the **Evidence Selection** module. In fact-checking, failing to retrieve a critical piece of "Gold Evidence" (a False Negative) is significantly more detrimental to the final verdict than including a small amount of irrelevant context (a False Positive).

To ensure an unbiased evaluation across all categories within each module, we report the Macro-average for all F_β scores. The Macro-average calculates the metric independently for each class and then takes the unweighted arithmetic mean:

$$F_{\beta,\text{ macro}} = \frac{1}{N}\sum_{i=1}^{N} F_{\beta,i} \tag{19}$$

where N represents the total number of classes or categories in the specific task. By consistently using the Macro-average format, we ensure that the model's performance on every category - from diverse news topics to different veracity labels - is reflected equally, preventing the final results from being skewed by imbalanced data distributions.

7.2. Experimental Results

This section presents the experimental results for the VietFactCheck system, beginning with a detailed performance analysis of each individual component: Claim Detection, Document Retrieval, Evidence Selection, and Claim Verification. Beyond evaluating these modules in isolation, we further examine their performance when integrated into complete, end-to-end pipelines. By showcasing the results of these combined configurations, we aim to provide deeper insights into the practical utility and robustness of our system when deployed in real-world information scenarios.

7.2.1. Claim Detection

The overall claim detection performance of both the Heuristic-based extraction module and the BERTSum-based neural model is summarized in Table 11. The experimental results clearly demonstrate that the BERTSum model substantially outperforms the heuristic baseline across all evaluated metrics and dataset splits. On the Development set, BERTSum achieves significant absolute gains, outperforming the heuristic method by 0.386 in Precision@1, 0.325 in Precision@3, and 0.239 in Precision@5.

A similar trend of improvement is observed on the Test set, where BERTSum elevates the Precision@1 from 0.620 to 0.907 and the Precision@5 from 0.413 to 0.668. This robust performance on unseen data highlights the model's strong generalization capabilities. The consistency of these results extends to the Training split, where BERTSum reaches a Precision@1 of 0.825 and a Precision@5 of 0.731, far exceeding the 0.473 and 0.438 recorded for the heuristic method. Collectively, these findings indicate that the Transformer-based model effectively captures deep semantic and contextual cues that go beyond the surface-level factual verb patterns utilized by the rule-based approach.

Despite the performance gap, the Heuristic approach maintains a reasonable level of efficacy, particularly considering its simplicity and the absence of supervised training. When viewed alongside our earlier analysis of factual verbs, these results suggest that statistically-derived linguistic signals offer a meaningful baseline for claim detection. More importantly, while BERTSum provides high-precision neural depth, the heuristic method offers an interpretable prior. This transparency makes the heuristic module a valuable asset that can complement neural architectures through initialization, pre-filtering, or by serving as a component within a hybrid fact-checking pipeline.

Methods	Dev			Test			Train		
	P@1	**P@3**	**P@5**	**P@1**	**P@3**	**P@5**	**P@1**	**P@3**	**P@5**
Heuristic	0.582	0.473	0.321	0.620	0.568	0.413	0.473	0.496	0.438
BERTSum	**0.968**	**0.798**	**0.560**	**0.907**	**0.893**	**0.668**	**0.825**	**0.830**	**0.731**

Table 11: Comparative performance of the Heuristic and BERTSum-based models for Claim Detection across Development, Test, and Training splits. P@k is the abbreviation of Precision@k.

7.2.2. Document Retrieval and Evidence Selection

The experimental results for the Document Retrieval and Evidence Selection stages are detailed in Table 12, illustrating a clear performance hierarchy between the two evaluated methodologies. Our findings highlight that the integration of a Reranking stage provides a substantial boost to retrieval quality. Specifically, the Document Retrieval Accuracy increases from 95.02% in the “Retrieval Only” baseline to 97.17% when Reranking is applied. This improvement extends to the Evidence Selection module, where the F_2 Score - a metric emphasizing recall—rises from 43.59% to 47.92%.

However, this elevation in precision introduces a significant computational trade-off in terms of processing latency. While the “Retrieval Only” method is highly efficient, processing a document sample in just 1.91 seconds, the “Retrieval + Reranking” pipeline requires 15.83 seconds - representing an approximately 8.3x increase in execution time. A similar trend is observed in the Evidence Selection phase, where time requirements escalate from a negligible 0.43 seconds to 6.69 seconds per sample.

These results suggest that while the reranking mechanism is essential for maximizing the reliability of the VietFactCheck system, its deployment in real-world, high-traffic scenarios necessitates careful optimization. For time-sensitive applications, the “Retrieval Only” approach remains a viable lightweight alternative, whereas the “Retrieval + Reranking” configuration serves as the gold standard for tasks requiring the highest possible evidence accuracy.

Methods	Document Retrieval		Evidence Selection	
	#Acc (%)	**#Time (s)**	**#F2 (%)**	**#Time (s)**
Retrieval Only	95.02	**1.91**	43.59	**0.43**
Retrieval + Reranking	**97.17**	15.83	**47.92**	6.69

Table 12: Comparative performance analysis of Document Retrieval and Evidence Selection methodologies on the ViFactCheck validation set. The evaluation measures Accuracy (#Acc) for document retrieval and the F_2 Score (#F2) for evidence selection. The Completion Time (#Time) metric denotes the average duration in seconds required to process a single sample from the ViFactCheck dataset for the respective module. Note: Performance measurements for #Time were conducted on a workstation equipped with an Intel Core i7-11850H CPU (2.50 GHz, 8 cores, 16 logical processors), 32 GB of RAM, and an NVIDIA RTX A2000 Laptop GPU. The Retrieval + Reranking method of these two modules was measured with the top-k value of 3 and 10, respectively.

7.2.3. Claim Verification

The results in Table 13 provide a clear hierarchy of model performance, highlighting the efficacy of large-scale pre-trained encoders in handling the semantic complexities of Vietnamese claim verification. Consistent with the original findings, the XLM-R$_{large}$ model emerged as the top performer in our experiments, achieving the highest accuracy of 68.52% in the Full Context setting and 84.27% when provided with Gold Evidence. This validates XLM-R$_{large}$ as a robust backbone for verifying claims that require deep linguistic understanding and cross-lingual knowledge transfer.

A significant observation across all evaluated models is the substantial performance gain achieved in the Gold Evidence scenario compared to the Full Context setting. For instance, our XLM-R$_{large}$ implementation sees an absolute accuracy increase of approximately 15.75% when the search space is restricted to specific evidence sentences. This trend underscores a critical bottleneck in the fact-checking pipeline: even the most powerful verification models struggle with "noise" from non-relevant sentences within a full context. This further emphasizes that the overall success of the VietFactCheck system is heavily dependent on the precision of the preceding Evidence Selection module.

In comparing our results to those established in the original paper, our implementation shows high fidelity to the established trends. While our scores for XLM-R$_{large}$ are slightly more conservative than the paper's reported 75.42% and 88.02%, we achieved a notably higher score for PhoBERT$_{large}$ in the Full Context setting (65.80% vs 62.93%). These minor variances are likely due to differences in training hyperparameters or fine-tuning schedules. Nevertheless, the consistency in the relative ranking of models - where large-scale multilingual and Vietnamese-optimized models consistently outperform smaller architectures like mBERT or ViBERT - confirms the reliability of the proposed verification framework.

Models	Our Result		Vifactcheck's Result	
	Full Context	Gold Evidence	Full Context	Gold Evidence
PhoBERT$_{base}$	62.30	78.94	<u>68.55</u>	77.76
PhoBERT$_{large}$	<u>65.80</u>	79.67	62.93	79.76
ViBERT	24.09	61.20	59.95	72.18
mBERT	41.09	73.18	58.07	69.94
XLM-R$_{base}$	61.54	<u>80.27</u>	65.40	<u>81.10</u>
XLM-R$_{large}$	**68.52**	**84.27**	**75.42**	**88.02**

Table 13: Comparative performance of various transformer-based encoders on the Claim Verification task using the ViFactCheck validation set. The results evaluate model accuracy across two settings: Full Context (inferring from the entire document) and Gold Evidence (inferring from isolated, ground-truth evidence). Our findings are benchmarked against the original results from the ViFactCheck paper to validate the replication and consistency of the experimental framework.

7.2.4. Combined Modules

To provide a clearer understanding of the system's practical performance, we evaluate several Combined Modules configurations. This involves integrating specific stages into end-to-end pipelines to observe how improvements in individual components propagate throughout the VietFactCheck architecture. For all pipelines, the top-k value was configured of 3 with the Document Retrieval module and 10 with the Evidence Selection one.

Document Retrieval + Evidence Selection The experimental results for the integrated retrieval and selection pipeline are detailed in Table 14, illustrating how different combinations of "Retrieval Only" and "Retrieval + Reranking" affect the final evidence identification quality. The data demonstrates that applying Reranking to either stage consistently yields a higher F_2 Score compared to the baseline configuration of 41.80%, where both modules utilize simple retrieval. Notably, the performance gains are significantly more substantial when Reranking is implemented at the Evidence Selection stage, pushing the score to 46.12% even without an advanced document retrieval step. The peak performance of 47.78% is achieved when both modules utilize the Reranking mechanism, confirming that the cumulative precision from both stages is vital for maximizing the system's overall effectiveness in isolating relevant facts.

Document Retrieval	Evidence Selection	#F2
Retrieval Only	Retrieval Only	41.80
Retrieval + Reranking	Retrieval Only	42.68
Retrieval Only	Retrieval + Reranking	<u>46.12</u>
Retrieval + Reranking	Retrieval + Reranking	**47.78**

Table 14: Combined performance evaluation of integrated Document Retrieval and Evidence Selection pipelines on the ViFactCheck validation set. The results highlight the synergistic impact of incorporating reranking mechanisms across both stages, measured by the final F_2 Score (#F2).

Retrieval Modules + Claim Verification The end-to-end verification results in Table 15 show that XLM-R$_{large}$ remains the superior model, achieving a peak accuracy of 74.52% when provided with selected evidence. Generally, isolating specific evidence segments improves performance by reducing "noise" compared to the Full Context setting; for example, XLM-R$_{large}$ improves from 68.01% to 73.87% in the RO+RO configuration. However, incorporating Reranking (RR) into the Evidence Selection stage presents a critical performance risk, as accuracy drops to 66.64% in the RR+RR setting. This suggests that while RR might optimize statistical retrieval metrics like F_2, it may inadvertently select segments that are less semantically aligned with the classifier's needs. When considering the 8x increase in processing time previously identified, the use of RR in a combined pipeline requires a careful trade-off between computational cost and final verification reliability.

Models	Document Retrieval		Document Retrieval + Evidence Selection			
	RO	RR	RO + RO	RR + RO	RO + RR	RR + RR
PhoBERT$_{base}$	61.81	62,28	63.05	63.60	60.49	60.80
PhoBERT$_{large}$	<u>65.21</u>	<u>65.68</u>	<u>70.25</u>	<u>70.94</u>	<u>64.76</u>	<u>65.32</u>
ViBERT	24.53	24.55	46.98	47.08	47.90	47.76
mBERT	40.85	40.81	64.52	64.55	59.29	59.76
XLM-R$_{base}$	61.31	61.10	66.63	66.78	62.56	62.24
XLM-R$_{large}$	**68.01**	**68.32**	**73.87**	**74.52**	**66.24**	**66.64**

Table 15: Verification accuracy across integrated modules on the ViFactCheck validation set. The table contrasts the Document Retrieval scenario (Full Context) against the Document Retrieval + Evidence Selection scenario (Selected Evidence). Abbreviations RO and RR denote Retrieval Only and Retrieval + Reranking, respectively.

7.3. Discussion

In this section, we provide a detailed analysis of the key factors influencing the performance of the VietFactCheck system. We examine the impact of context length, evidence quantity, and top-k values on the efficiency of our Document Retrieval, Evidence Selection, and Claim Verification modules. Additionally, we discuss the role of factual verbs in determining the effectiveness of our heuristic-based Claim Detection approach, providing insights into its operational strengths and limitations.

7.3.1. Impact of Factual Verbs Statistics on Heuristic-based Claim Detection

The effectiveness of our heuristic-based approach is fundamentally rooted in the identification of stable linguistic markers. Table 16 catalogs the primary “factual verbs” that exhibit high frequency and consistent presence across all dataset splits. Verbs such as “có”, “được”, and “là” not only appear thousands of times but also maintain a steady presence across the Train, Dev, and Test sets, suggesting they serve as reliable indicators of verifiable statements in the Vietnamese language.

Verb	Train	Dev	Test	Total	Presence	Mean
có	1471	118	279	1868	3	622.67
được	1153	93	298	1544	3	514.67
là	1017	69	210	1296	3	432.00
ra	561	44	117	722	3	240.67
bị	542	45	112	699	3	233.00
theo	448	36	114	598	3	199.33
cho	377	22	89	488	3	162.67
phải	312	34	65	411	3	137.00
làm	281	20	75	376	3	125.33
tổ chức	281	23	68	372	3	124.00
thực hiện	251	18	41	310	3	103.33
phát triển	232	16	48	296	3	98.67
cần	211	18	47	276	3	92.00
lên	197	21	45	263	3	87.67
biết	244	0	62	306	2	102.00
sử dụng	214	0	46	260	2	86.67
đi	183	0	44	227	2	75.67
gây	0	17	41	58	2	19.33
thi	219	0	0	219	1	73.00
lại	204	0	0	204	1	68.00
tăng	196	0	0	196	1	65.33
xây dựng	0	0	43	43	1	14.33
tạo	0	0	42	42	1	14.00
quản lý	0	19	0	19	1	6.33
nhận	0	18	0	18	1	6.00
đầu tư	0	18	0	18	1	6.00
yêu cầu	0	17	0	17	1	5.67
hoạt động	0	15	0	15	1	5.00

Table 16: Statistical summary of consensus factual verbs identified across Training, Development, and Test splits.

The quantitative advantage of utilizing these corpus-level statistics is clearly demonstrated in Table 17. When comparing our Statistics-based selection to a Random sampling baseline, the former consistently achieves superior or comparable performance, particularly as the ranking depth increases. On the Development set, our approach improves Precision@3 (0.473 vs. 0.459) and Precision@5 (0.321 vs. 0.306), indicating a higher overall quality in ranking potential claims.

This trend is even more pronounced on the Test set, where the statistics-based method yields significant absolute gains in Precision@1 (0.620 vs. 0.570), Precision@3 (0.568 vs. 0.525), and Precision@5 (0.413 vs. 0.379). The fact that the performance gap widens on unseen data validates our chosen factual verb inventory as highly generalizable and robust against dataset-specific artifacts.

While the Random baseline occasionally matches the performance at Precision@1 (as seen in the Train and Dev splits), it fails to maintain stability across wider evaluation ranges. In contrast, the statistics-driven strategy dominates medium- and long-range precision, which is critical for practical claim detection scenarios where multiple candidate sentences must be evaluated. These findings confirm that using frequency and split presence to construct a factual verb set provides a principled and interpretable foundation for heuristic identification, effectively capturing the structural essence of factual discourse.

Methods	**Dev**			**Test**			**Train**		
	P@1	**P@3**	**P@5**	**P@1**	**P@3**	**P@5**	**P@1**	**P@3**	**P@5**
Statistics based	0.582	**0.473**	**0.321**	**0.620**	**0.568**	**0.413**	0.473	**0.496**	**0.438**
Random	**0.585**	0.459	0.306	0.570	0.525	0.379	**0.474**	0.492	0.424

Table 17: Comparative impact analysis of factual verb selection strategies on heuristic-based claim detection. The results contrast the performance of a Statistics-based selection (using the inventory from Table 16) against a Random sampling baseline, measured by Precision@1, Precision@3, and Precision@5 across all dataset splits.

7.3.2. Impact of Full Context Length on Claim Verification

The impact of context length on the claim verification task is illustrated in Figure 10, revealing distinct performance patterns among the evaluated models. A primary observation is the significant and consistent performance gap between large-scale encoders, specifically XLM-R_{large} and PhoBERT$_{large}$, and their base or multilingual counterparts like mBERT and ViBERT across all token ranges. These high-capacity models demonstrate remarkable resilience to semantic noise, maintaining relatively stable F1 Macro Scores even as the context length extends beyond 1,000 tokens. Notably, XLM-R_{large} exhibits near-perfect performance at short context lengths (0-100 tokens) and retains its leadership even in extremely long documents (>1500 tokens), whereas models like mBERT and ViBERT consistently struggle to extract meaningful features amidst increasing textual density.

This stability - particularly within the 200 to 1,000-token range - suggests that the attention mechanisms in these large-scale architectures effectively prioritize relevant evidence even when buried in extraneous context. Such findings reinforce the selection of XLM-R_{large} as the optimal backbone for the VietFactCheck pipeline, ensuring reliable verification across the diverse lengths typically found in online Vietnamese news reports.

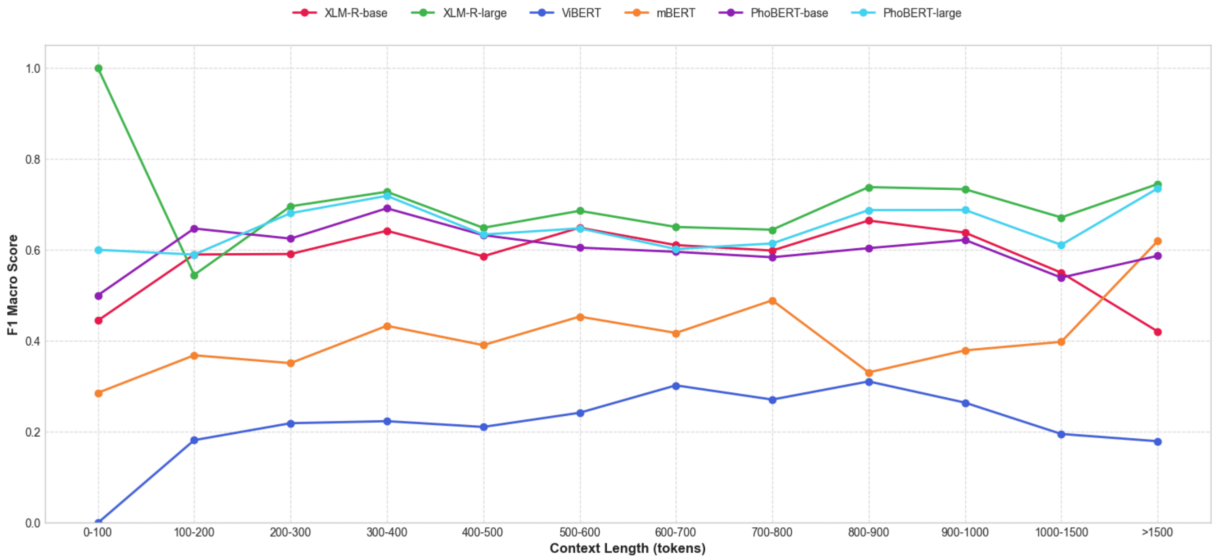


Figure 10: Comparative analysis of verification performance (F1 Macro Score) across varying context lengths (tokens) for diverse transformer-based encoders. The chart illustrates the models' resilience to semantic noise as the volume of information within the full context increases.

7.3.3. Impact of Evidence Quantity on Claim Verification

The results presented in Figure 11 reveal a consistent trend where verification accuracy is highest when the evidence provided is limited to a concise set of one or two sentences. XLM-R$_{large}$ achieves its peak performance of approximately 0.9 in the two-evidence scenario, outperforming all other models. However, as the number of evidence sentences increases to three and beyond, a significant decline in F1 Macro Score is observed across nearly all architectures. This performance drop suggests that while multiple evidences may aim for thoroughness, they often introduce semantic noise or conflicting information that distracts the verification model. These findings emphasize that for the VietFactCheck pipeline, the precision of the Evidence Selection stage is more critical than its recall; providing a small, high-quality set of relevant evidence is essential to avoid the accuracy degradation caused by information overload.

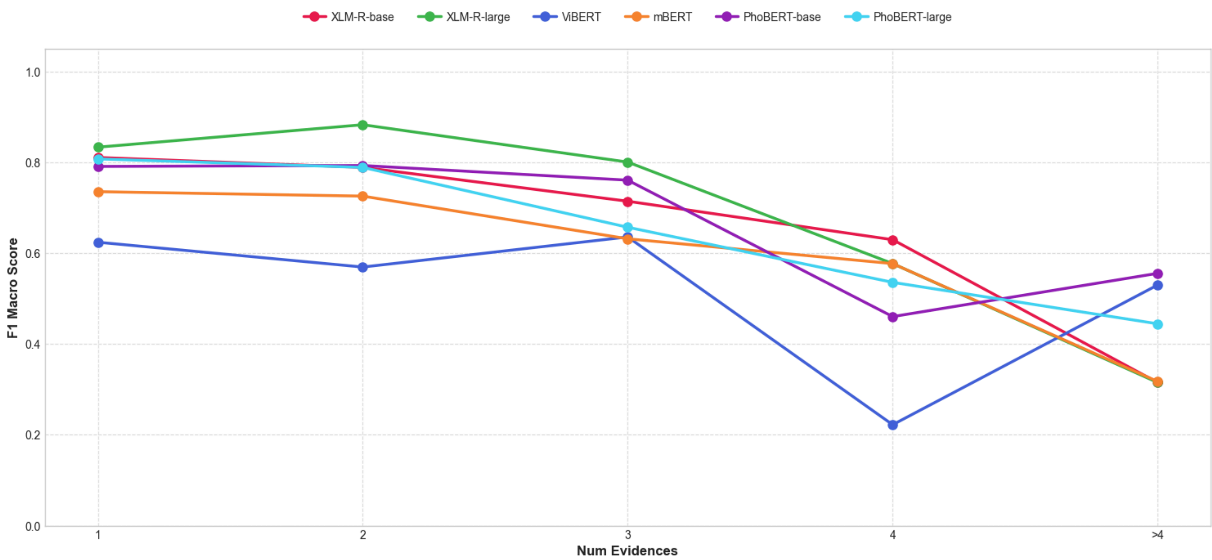


Figure 11: Verification performance (F1 Macro Score) across different quantities of retrieved evidence sentences. The chart demonstrates the relationship between the number of evidence pieces provided to the classifier and the resulting accuracy of the final verdict.

7.3.4. Impact of Top-k Value on Retrieval Modules

In the Evidence Selection task (Figure 12), the Retrieval + Reranking (RR) configuration consistently and significantly outperforms the Retrieval Only (RO) baseline across every evaluated Top-k value. While the performance of the RO method steadily declines as k increases—likely due to the inclusion of more non-relevant sentences—the RR strategy maintains a high and stable Macro F2 Score, proving its robustness in identifying true evidence within larger candidate pools.

Simultaneously, the Document Retrieval performance shown in Figure 13 demonstrates that accuracy initially follows a steady upward trend as the Top-k value grows, reaching its peak effectiveness at approximately Top-k = 5. Beyond this point, however, the accuracy plateaus and begins to show a gradual decline. This behavior suggests that while a Top-k of 5 provides sufficient coverage to capture the correct source document, higher values introduce excessive semantic noise that can mislead the retrieval process. Consequently, these results validate our selection of Top-k = 5 as the optimal configuration for balancing high retrieval accuracy with computational efficiency within the end-to-end pipeline.

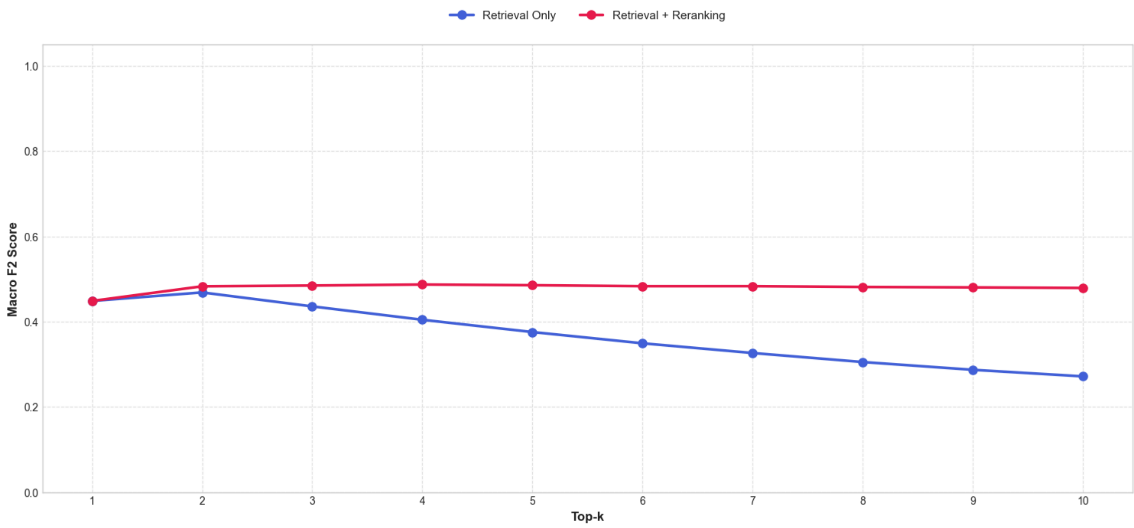


Figure 12: Impact of Top-k values on the Macro F2 Score for Evidence Selection. The chart contrasts the performance of the Retrieval Only baseline against the Retrieval + Reranking strategy.

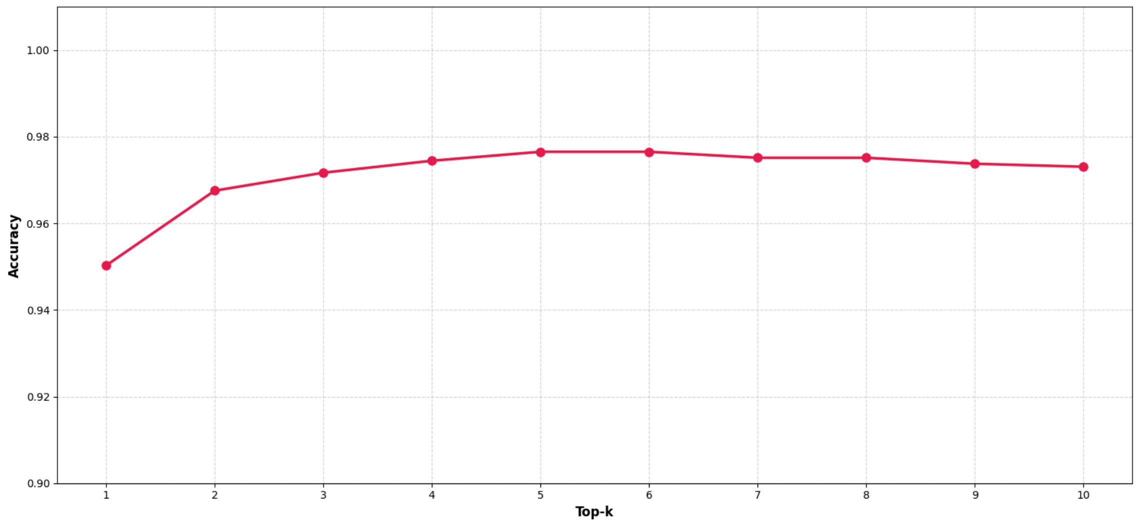


Figure 13: Accuracy of the Document Retrieval module across varying Top-k values. This visualization illustrates the threshold at which increasing the number of retrieved candidates ceases to improve document identification accuracy.

8. Our Website Application

The VietFactCheck web application serves as the practical implementation and demonstration interface for the multi-stage fact-checking pipeline proposed in this study. By integrating the specialized modules for claim detection, document retrieval, and verification, the application provides a tangible platform for real-time news assessment and system evaluation. This section details the system's deployment through two primary lenses: the Overview, which outlines the general design and user interface philosophy, and the Core Functionality, which describes the specific technical capabilities and interactive features embedded within the platform.

8.1. Overview

The VietFactCheck web application is designed as an interactive environment for transparent, step-by-step news verification. Utilizing a dual-pane layout, the interface features a centralized sidebar that allows users to adjust technical parameters - such as hybrid search weighting and Vietnamese language model selection - in real time. This configuration-centric design (illustrated in Figure 14) ensures that the system's logic remains accessible, transforming the application from a static tool into a dynamic research environment.

The user journey starts with a multi-modal recommendation engine that categorizes news into a responsive grid, facilitating exploration across diverse linguistic contexts. Once a story is selected, the interface transitions to specific claim analysis through a dynamic tabbed system. This approach isolates individual assertions within complex articles, ensuring each claim is scrutinized against targeted evidence rather than just providing a generic verdict.

To maintain total transparency, the application maps retrieved data back to its original source via contextual highlighting. This method preserves the integrity of the source document while pinpointing the exact evidence used for verification. By incorporating state-aware indicators to track the processing pipeline, the application provides a clear temporal map of the task, ensuring that every final classification is grounded in visible and verifiable evidence.

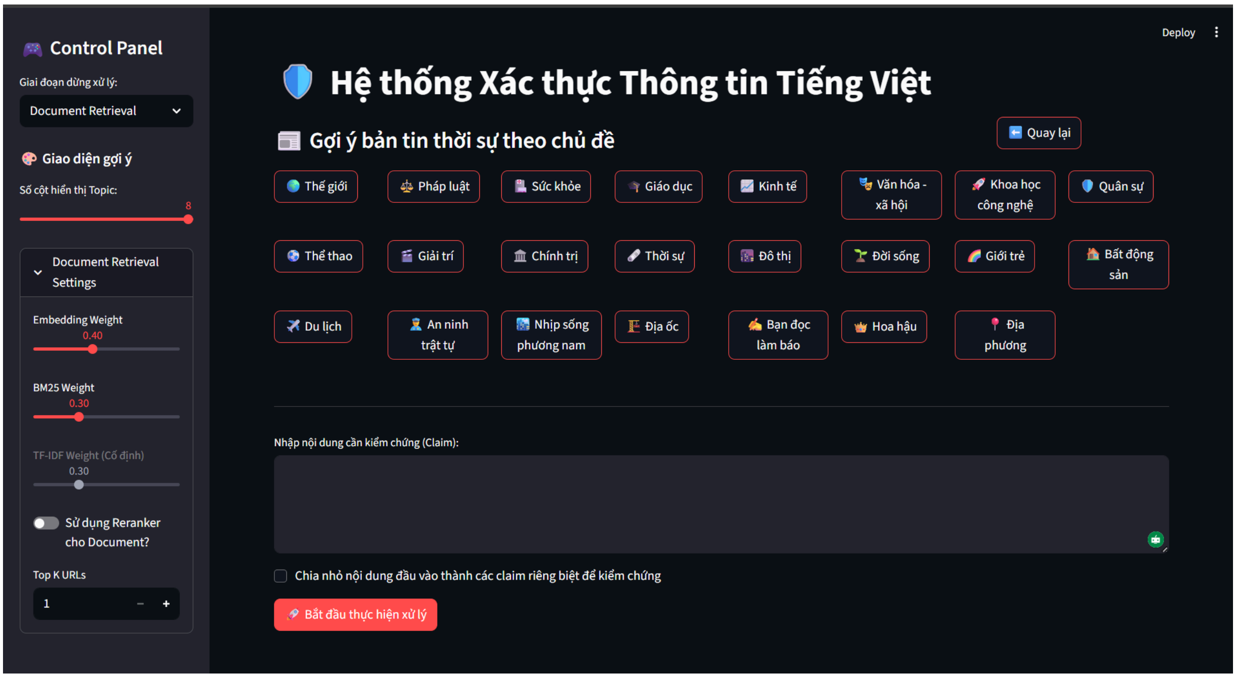


Figure 14: The VietFactCheck web interface, featuring a dual-pane layout that integrates real-time pipeline configuration (left) with a multi-modal news recommendation and claim submission dashboard (right).

8.2. Core Functionality

The core functionality of the VietFactCheck application is designed to provide a comprehensive and modular environment for verifying the authenticity of news content. By integrating advanced natural language processing techniques into an interactive interface, the system ensures that every stage of the fact-checking process - from initial exploration to final classification - is both transparent and configurable. The following sections describe the key capabilities that define the application's operational logic and user experience.

Dynamic Topic-based Recommendation Engine This feature provides users with an immediate point of entry by loading and shuffling verified claims and news context from the system's training datasets. The engine populates a grid of interactive topic buttons - ach identified by a unique icon ranging from "Chính trị" (Politics) to "Số hóa" (Digitalization) - that allow for the rapid selection of subject matter across 36 distinct categories. This mechanism is supported by a state management system that enables users to switch between "Claim mode" for short assertions and "News mode" for multi-sentence contexts, effectively demonstrating the system's ability to handle diverse linguistic structures before a single word is manually typed.

Automated Claim Extraction and Decomposition The application utilizes a specialized BERTSum module to address the inherent complexity of unstructured news articles. This component analyzes a long block of text to identify and extract "atomic claims" - the specific, verifiable assertions that require checking. Once extracted, the application dynamically generates a tab-based navigation system, isolating each individual claim so that it can be processed through the subsequent retrieval and verification modules independently. This approach ensures that a single news report containing a mixture of accurate and inaccurate information is evaluated with high resolution, providing specific verdicts for each statement rather than a single, ambiguous label for the entire document.

Adaptive Multistage Pipeline Control The system offers a unique level of modularity by allowing users to define the terminal point of the fact-checking process through a dedicated "Control Panel". Users can instruct the backend to halt processing after "Document Retrieval," "Evidence Selection," or "Claim Verification," which significantly aids in debugging and system transparency. When a specific stage is selected, the application adaptively executes only the prerequisite modules - such as identifying top-K URLs or filtering for relevant evidence sentences - and renders the intermediate outputs directly. This allows researchers to evaluate the quality of the search results or the relevance of selected evidence snippets in isolation without waiting for the final classification model to run.

Interactive Parameter Tuning and Benchmarking Our platform provides a centralized configuration suite where users can manipulate the fundamental mathematical weights governing the retrieval modules in real-time. This includes independent sliders to adjust the influence of semantic embeddings, BM25, and TF-IDF for both document and evidence retrieval, supported by UI safeguards that prevent invalid weight totals. Additionally, the system allows for the benchmarking of various fine-tuned Vietnamese Pre-trained Language Models (PLMs), such as PhoBERT$_{large}$, XLM-RoBERTa, and ViBERT, alongside adjustable reranking thresholds (T_1 for confidence and T_2 for score gaps). This capability transforms the verification process into a "glass-box" experiment, where the user can directly observe how specific model choices and retrieval settings affect the final determination of "Supported," "Refuted," or "Unknown".

9. Conclusion and Future Work

In this project, we have successfully developed VietFactCheck, a comprehensive end-to-end fact-checking system designed for the Vietnamese language. While previous research on the ViFactCheck dataset focused extensively on the verification stage , our work bridges the gap by implementing and evaluating the full pipeline: Claim Detection, Document Retrieval, Evidence Selection, and Claim Verification. Our technical experiments provided a deeper dive into the retrieval process, utilizing a hybrid combination of SBERT embeddings, BM25, and TF-IDF. We demonstrated that the integration of a Cross-Encoder Reranker significantly enhances the quality of evidence retrieved compared to standard methods, which is crucial for handling complex inferential chains. To ensure practical usability, the verification module was optimized for personal computing environments using fine-tuned Pre-trained Language Models (PLMs) like PhoBERT and mBERT, achieving robust performance as measured by Accuracy, Macro F1-score, and Macro F2-score. Furthermore, we contributed a new synthetic dataset consisting of fake contexts generated from original claims to specifically benchmark and refine the Claim Extraction module.

Looking ahead, we aim to enhance the transparency and scope of the system through several key advancements. Our primary goal is to integrate an explainability mechanism that provides natural language justifications for the model's verdicts, helping users understand the rationale behind each "Supported" or "Refuted" label. We also plan to explore the integration of Large Language Models (LLMs) to handle more nuanced reasoning, while investigating deployment strategies—such as quantization or specialized APIs—to mitigate the latency issues typically associated with these larger architectures. By addressing identified challenges such as semantic ambiguity and evidence retrieval failures, we seek to create a more resilient tool capable of accurately navigating the complex digital media landscape in Vietnam. These efforts will pave the way for a more reliable and user-centric approach to combating misinformation in low-resource linguistic environments.

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